Chapter 12

Linear Bayesian Estimators

12.1 Introduction

The optimal Bayesian estimators discussed in the previous chapter are difficult to determine in closed form, and in practice too computationally intensive to implement. They involve multidimensional integration for the MMSE estimator and multidimensional maximization for the MAP estimator. Although under the jointly Gaussian assumption these estimators are easily found, in general, they are not. When we are unable to make the Gaussian assumption, another approach must be used. To fill this gap we can choose to retain the MMSE criterion but constrain the estimator to be *linear*. Then, an explicit form for the estimator may be determined which depends only on the first two moments of the PDF. In many ways this approach is analogous to the BLUE in classical estimation, and some parallels will become apparent. In practice, this class of estimators, which are generically termed *Wiener filters*, are extensively utilized.

12.2 Summary

The linear estimator is defined by (12.1), and the corresponding Bayesian MSE by (12.2). Minimizing the Bayesian MSE results in the linear MMSE (LMMSE) estimator of (12.6) and the minimum Bayesian MSE of (12.8). The estimator may also be derived using a vector space viewpoint as described in Section 12.4. This approach leads to the important orthogonality principle which says that the error of the linear MMSE estimator must be uncorrelated with the data. The vector LMMSE estimator is given by (12.20), and the minimum Bayesian MSE by (12.21) and (12.22). The estimator commutes over linear transformations (12.23) and has an additivity property (12.24). For a data vector having the Bayesian linear model form (the Bayesian linear model without the Gaussian assumption), the LMMSE estimator and its performance are summarized in Theorem 12.1. If desired, the estimator for the Bayesian linear model form can be implemented sequentially in time using (12.47)–(12.49).

12.3 Linear MMSE Estimation

We begin our discussion by assuming a scalar parameter θ is to be estimated based on the data set $\{x[0], x[1], \ldots, x[N-1]\}$ or in vector form $\mathbf{x} = [x[0]\, x[1] \ldots x[N-1]]^T$. The unknown parameter is modeled as the realization of a random variable. We do not assume any specific form for the joint PDF $p(\mathbf{x}, \theta)$, but as we shall see shortly, only a knowledge of the first two moments. That θ may be estimated from $\mathbf{x}$ is due to the assumed statistical dependence of θ on $\mathbf{x}$ as summarized by the joint PDF $p(\mathbf{x}, \theta)$, and in particular, for a linear estimator we rely on the correlation between θ and $\mathbf{x}$. We now consider the class of all linear (actually affine) estimators of the form

$$\hat{\theta} = \sum_{n=0}^{N-1} a_n x[n] + a_N \tag{12.1}$$

and choose the weighting coefficients a_n's to minimize the Bayesian MSE

$$\text{Bmse}(\hat{\theta}) = E\left[(\theta - \hat{\theta})^2\right] \tag{12.2}$$

where the expectation is with respect to the PDF $p(\mathbf{x}, \theta)$. The resultant estimator is termed the *linear minimum mean square error (LMMSE) estimator*. Note that we have included the a_N coefficient to allow for nonzero means of $\mathbf{x}$ and θ. If the means are both zero, then this coefficient may be omitted, as will be shown later.

Before determining the LMMSE estimator we should keep in mind that the estimator will be suboptimal unless the MMSE estimator happens to be linear. Such would be the case, for example, if the Bayesian linear model applied (see Section 10.6). Otherwise, better estimators will exist, although they will be nonlinear (see the introductory example in Section 10.3). Since the LMMSE estimator relies on the correlation between random variables, a parameter uncorrelated with the data cannot be linearly estimated. Consequently, the proposed approach is not always feasible. This is illustrated by the following example. Consider a parameter θ to be estimated based on the single data sample $x[0]$, where $x[0] \sim \mathcal{N}(0, \sigma^2)$. If the parameter to be estimated is the power of the $x[0]$ *realization* or $\theta = x^2[0]$, then a perfect estimator will be

$$\hat{\theta} = x^2[0]$$

since the minimum Bayesian MSE will be zero. This estimator is clearly nonlinear. If, however, we attempt to use a LMMSE estimator or

$$\hat{\theta} = a_0 x[0] + a_1,$$

then the optimal weighting coefficients a_0 and a_1 can be found by minimizing

$$\begin{aligned} \text{Bmse}(\hat{\theta}) &= E\left[(\theta - \hat{\theta})^2\right] \\ &= E\left[(\theta - a_0 x[0] - a_1)^2\right]. \end{aligned}$$

We differentiate this with respect to a_0 and a_1 and set the results equal to zero to produce

$$\begin{aligned} E\left[(\theta - a_0 x[0] - a_1)x[0]\right] &= 0 \\ E(\theta - a_0 x[0] - a_1) &= 0 \end{aligned}$$

or

$$\begin{aligned} a_0 E(x^2[0]) + a_1 E(x[0]) &= E(\theta x[0]) \\ a_0 E(x[0]) + a_1 &= E(\theta). \end{aligned}$$

But $E(x[0]) = 0$ and $E(\theta x[0]) = E(x^3[0]) = 0$, so that

$$\begin{aligned} a_0 &= 0 \\ a_1 &= E(\theta) = E(x^2[0]) = \sigma^2. \end{aligned}$$

Therefore, the LMMSE estimator is $\hat{\theta} = \sigma^2$ and does not depend on the data. This is because θ and $x[0]$ are uncorrelated. The minimum MSE is

$$\begin{aligned} \text{Bmse}(\hat{\theta}) &= E\left[(\theta - \hat{\theta})^2\right] \\ &= E\left[(\theta - \sigma^2)^2\right] \\ &= E\left[(x^2[0] - \sigma^2)^2\right] \\ &= E(x^4[0]) - 2\sigma^2 E(x^2[0]) + \sigma^4 \\ &= 3\sigma^4 - 2\sigma^4 + \sigma^4 \\ &= 2\sigma^4 \end{aligned}$$

as opposed to a minimum MSE of zero for the nonlinear estimator $\theta = x^2[0]$. Clearly, the LMMSE estimator is inappropriate for this problem. Problem 12.1 explores how to modify the LMMSE estimator to make it applicable.

We now derive the optimal weighting coefficients for use in (12.1). Substituting (12.1) into (12.2) and differentiating

$$\frac{\partial}{\partial a_N} E\left[\left(\theta - \sum_{n=0}^{N-1} a_n x[n] - a_N\right)^2\right] = -2E\left[\theta - \sum_{n=0}^{N-1} a_n x[n] - a_N\right].$$

Setting this equal to zero produces

$$a_N = E(\theta) - \sum_{n=0}^{N-1} a_n E(x[n]) \tag{12.3}$$

which as asserted earlier is zero if the means are zero. Continuing, we need to minimize

$$\text{Bmse}(\hat{\theta}) = E\left\{\left[\sum_{n=0}^{N-1} a_n (x[n] - E(x[n])) - (\theta - E(\theta))\right]^2\right\}$$

over the remaining a_n's, where a_N has been replaced by (12.3). Letting $\mathbf{a} = [a_0\, a_1 \ldots a_{N-1}]^T$, we have

$$\begin{aligned}
\text{Bmse}(\hat{\theta}) &= E\left\{\left[\mathbf{a}^T(\mathbf{x} - E(\mathbf{x})) - (\theta - E(\theta))\right]^2\right\} \\
&= E\left[\mathbf{a}^T(\mathbf{x} - E(\mathbf{x}))(\mathbf{x} - E(\mathbf{x}))^T\mathbf{a}\right] - E\left[\mathbf{a}^T(\mathbf{x} - E(\mathbf{x}))(\theta - E(\theta))\right] \\
&\quad - E\left[(\theta - E(\theta))(\mathbf{x} - E(\mathbf{x}))^T\mathbf{a}\right] + E\left[(\theta - E(\theta))^2\right] \\
&= \mathbf{a}^T\mathbf{C}_{xx}\mathbf{a} - \mathbf{a}^T\mathbf{C}_{x\theta} - \mathbf{C}_{\theta x}\mathbf{a} + C_{\theta\theta} \qquad (12.4)
\end{aligned}$$

where $\mathbf{C}_{xx}$ is the $N \times N$ covariance matrix of $\mathbf{x}$, and $\mathbf{C}_{\theta x}$ is the $1 \times N$ cross-covariance vector having the property that $\mathbf{C}_{\theta x}^T = \mathbf{C}_{x\theta}$, and $C_{\theta\theta}$ is the variance of θ. Making use of (4.3) we can minimize (12.4) by taking the gradient to yield

$$\frac{\partial \text{Bmse}(\hat{\theta})}{\partial \mathbf{a}} = 2\mathbf{C}_{xx}\mathbf{a} - 2\mathbf{C}_{x\theta}$$

which when set to zero results in

$$\mathbf{a} = \mathbf{C}_{xx}^{-1}\mathbf{C}_{x\theta}. \qquad (12.5)$$

Using (12.3) and (12.5) in (12.1) produces

$$\begin{aligned}
\hat{\theta} &= \mathbf{a}^T\mathbf{x} + a_N \\
&= \mathbf{C}_{x\theta}^T\mathbf{C}_{xx}^{-1}\mathbf{x} + E(\theta) - \mathbf{C}_{x\theta}^T\mathbf{C}_{xx}^{-1}E(\mathbf{x})
\end{aligned}$$

or finally the *LMMSE estimator* is

$$\hat{\theta} = E(\theta) + \mathbf{C}_{\theta x}\mathbf{C}_{xx}^{-1}(\mathbf{x} - E(\mathbf{x})). \qquad (12.6)$$

Note that it is identical in form to the MMSE estimator for jointly Gaussian $\mathbf{x}$ and θ, as can be verified from (10.24). This is because in the Gaussian case the MMSE estimator happens to be linear, and hence our constraint is automatically satisfied. If the means of θ and $\mathbf{x}$ are zero, then

$$\hat{\theta} = \mathbf{C}_{\theta x}\mathbf{C}_{xx}^{-1}\mathbf{x}. \qquad (12.7)$$

The minimum Bayesian MSE is obtained by substituting (12.5) into (12.4) to yield

$$\begin{aligned}
\text{Bmse}(\hat{\theta}) &= \mathbf{C}_{x\theta}^T\mathbf{C}_{xx}^{-1}\mathbf{C}_{xx}\mathbf{C}_{xx}^{-1}\mathbf{C}_{x\theta} - \mathbf{C}_{x\theta}^T\mathbf{C}_{xx}^{-1}\mathbf{C}_{x\theta} \\
&\quad - \mathbf{C}_{\theta x}\mathbf{C}_{xx}^{-1}\mathbf{C}_{x\theta} + C_{\theta\theta} \\
&= \mathbf{C}_{\theta x}\mathbf{C}_{xx}^{-1}\mathbf{C}_{x\theta} - 2\mathbf{C}_{\theta x}\mathbf{C}_{xx}^{-1}\mathbf{C}_{x\theta} + C_{\theta\theta}
\end{aligned}$$

or finally

$$\text{Bmse}(\hat{\theta}) = C_{\theta\theta} - \mathbf{C}_{\theta x}\mathbf{C}_{xx}^{-1}\mathbf{C}_{x\theta}. \qquad (12.8)$$

Again this is identical to that obtained by substituting (10.25) into (11.12). An example follows.

Example 12.1 - DC Level in WGN with Uniform Prior PDF

Consider the introductory example in Chapter 10. The data model is

$$x[n] = A + w[n] \qquad n = 0, 1, \ldots, N-1$$

where $A \sim \mathcal{U}[-A_0, A_0]$, $w[n]$ is WGN with variance σ^2, and A and $w[n]$ are independent. We wish to estimate A. The MMSE estimator cannot be obtained in closed form due to the integration required (see (10.9)). Applying the LMMSE estimator, we first note that $E(A) = 0$, and hence $E(x[n]) = 0$. Since $E(\mathbf{x}) = \mathbf{0}$, the covariances are

$$\begin{aligned} \mathbf{C}_{xx} &= E(\mathbf{x}\mathbf{x}^T) \\ &= E\left[(A\mathbf{1} + \mathbf{w})(A\mathbf{1} + \mathbf{w})^T\right] \\ &= E(A^2)\mathbf{1}\mathbf{1}^T + \sigma^2\mathbf{I} \\ \mathbf{C}_{\theta x} &= E(A\mathbf{x}^T) \\ &= E\left[A(A\mathbf{1} + \mathbf{w})^T\right] \\ &= E(A^2)\mathbf{1}^T \end{aligned}$$

where $\mathbf{1}$ is an $N \times 1$ vector of all ones. Hence, from (12.7)

$$\begin{aligned} \hat{A} &= \mathbf{C}_{\theta x}\mathbf{C}_{xx}^{-1}\mathbf{x} \\ &= \sigma_A^2\mathbf{1}^T(\sigma_A^2\mathbf{1}\mathbf{1}^T + \sigma^2\mathbf{I})^{-1}\mathbf{x} \end{aligned}$$

where we have let $\sigma_A^2 = E(A^2)$. But the form of the estimator is identical to that encountered in Example 10.2 if we let $\mu_A = 0$, so that from (10.31)

$$\hat{A} = \frac{\sigma_A^2}{\sigma_A^2 + \frac{\sigma^2}{N}}\bar{x}.$$

Since $\sigma_A^2 = E(A^2) = (2A_0)^2/12 = A_0^2/3$, the LMMSE estimator of A is

$$\hat{A} = \frac{\frac{A_0^2}{3}}{\frac{A_0^2}{3} + \frac{\sigma^2}{N}}\bar{x}. \tag{12.9}$$

As opposed to the original MMSE estimator which required integration, we have obtained the LMMSE estimator in closed form. Also, note that we did not really need to know that A was uniformly distributed but only its mean and variance, or that $w[n]$ was Gaussian but only that it is white and its variance. Likewise, independence of A and $\mathbf{w}$ was not required, only that they were uncorrelated. In general, all that is required to determine the LMMSE estimator are the first two moments of $p(\mathbf{x}, \theta)$ or

$$\begin{bmatrix} E(\theta) \\ E(\mathbf{x}) \end{bmatrix}, \begin{bmatrix} C_{\theta\theta} & \mathbf{C}_{\theta x} \\ \mathbf{C}_{x\theta} & \mathbf{C}_{xx} \end{bmatrix}.$$

However, we must realize that the LMMSE of (12.9) will be *suboptimal* since it has been *constrained to be linear.* The optimal estimator for this problem is given by (10.9).

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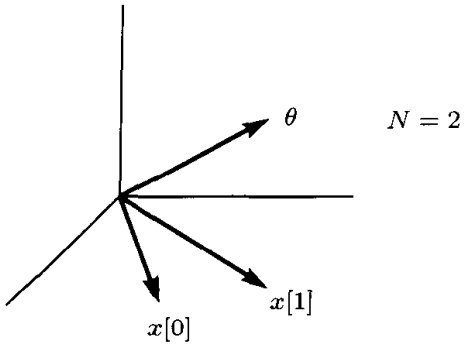


Figure 12.1 Vector space interpretation of random variables

12.4 Geometrical Interpretations

In Chapter 8 we discussed a geometrical interpretation of the LSE based on the concept of a vector space. The LMMSE estimator admits a similar interpretation, although now the "vectors" are random variables. (Actually, both vector spaces are special cases of the more general Hilbert space [Luenberger 1969].) This alternative formulation assumes that θ and $\mathbf{x}$ are zero mean. If they are not, we can always define the zero mean random variables $\theta' = \theta - E(\theta)$ and $\mathbf{x}' = \mathbf{x} - E(\mathbf{x})$, and consider estimation of θ' by a linear function of $\mathbf{x}'$ (see also Problem 12.5). Now we wish to find the a_n's so that

$$\hat{\theta} = \sum_{n=0}^{N-1} a_n x[n]$$

minimizes

$$\text{Bmse}(\hat{\theta}) = E\left[(\theta - \hat{\theta})^2\right].$$

Let us now think of the random variables $\theta, x[0], x[1], \ldots, x[N-1]$ as elements in a vector space as shown symbolically in Figure 12.1. The reader may wish to verify that the properties of a vector space are satisfied, such as vector addition and multiplication by a scalar, etc. Since, as is usually the case, θ cannot be perfectly expressed as a linear combination of the $x[n]$'s (if it could, then our estimator would be perfect), we picture θ as only partially lying in the subspace spanned by the $x[n]$'s. We may define the "length" of each vector x as $||x|| = \sqrt{E(x^2)}$ or the square root of the variance. Longer length vectors are those with larger variances. The zero length vector is the random variable with zero variance or, therefore, the one that is identically zero (actually not random at all). Finally, to complete our description we require the notion of an *inner product* between two vectors. (Recall that if $\mathbf{x}, \mathbf{y}$ are Euclidean vectors in R^3, then the inner product is $(\mathbf{x}, \mathbf{y}) = \mathbf{x}^T\mathbf{y} = ||\mathbf{x}||\,||\mathbf{y}||\cos\alpha$, where α is the angle between the vectors.) It can be shown that an appropriate definition, i.e., one that satisfies the properties of an inner product between the vectors x and y, is (see Problem 12.4)

$$(x, y) = E(xy). \tag{12.10}$$

With this definition we have that

$$(x, x) = E(x^2) = ||x||^2, \tag{12.11}$$

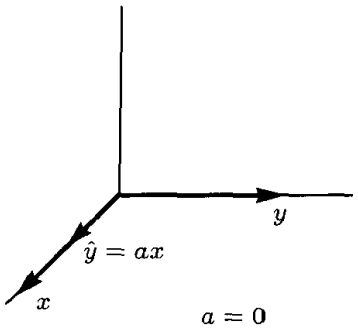


Figure 12.2 Orthogonal random variables—y cannot be linearly estimated based on x

consistent with our earlier definition of the length of a vector. Also, we can now define two vectors to be orthogonal if

$$(x, y) = E(xy) = 0. \tag{12.12}$$

Since the vectors are zero mean, this is equivalent to saying that two vectors are *orthogonal* if and only if they are *uncorrelated.* (In R^3 two Euclidean vectors are orthogonal if the angle between them is $\alpha = 90°$, so that $(\mathbf{x}, \mathbf{y}) = ||\mathbf{x}||\,||\mathbf{y}|| \cos\alpha = 0$.) Recalling our discussions from the previous section, this implies that if two vectors are orthogonal, we cannot use one to estimate the other. As shown in Figure 12.2, since there is no component of y along x, we cannot use x to estimate y. Attempting to do so means that we need to find an a so that $\hat{y} = ax$ minimizes the Bayesian MSE, $\text{Bmse}(\hat{y}) = E[(y - \hat{y})^2]$. To find the optimal value of a

$$\begin{aligned}
\frac{d\,\text{Bmse}(\hat{y})}{da} &= \frac{d}{da}E[(y - ax)^2] \\
&= \frac{d}{da}[E(y^2) - 2aE(xy) + a^2E(x^2)] \\
&= -2E(xy) + 2aE(x^2) \\
&= 0,
\end{aligned}$$

which yields

$$a = \frac{E(xy)}{E(x^2)} = 0.$$

The LMMSE estimator of y is just $\hat{y} = 0$. Of course, this is a special case of (12.7) where $N = 1$, $\theta = y$, $x[0] = x$, and $C_{\theta x} = 0$.

With these ideas in mind we proceed to determine the LMMSE estimator using the vector space viewpoint. This approach is useful for conceptualization of the LMMSE estimation process and will be used later to derive the sequential LMMSE estimator. As before, we assume that

$$\hat{\theta} = \sum_{n=0}^{N-1} a_n x[n]$$

where $a_N = 0$ due to the zero mean assumption. We wish to estimate θ as a linear combination of $x[0], x[1], \ldots, x[N-1]$. The weighting coefficients a_n should be chosen

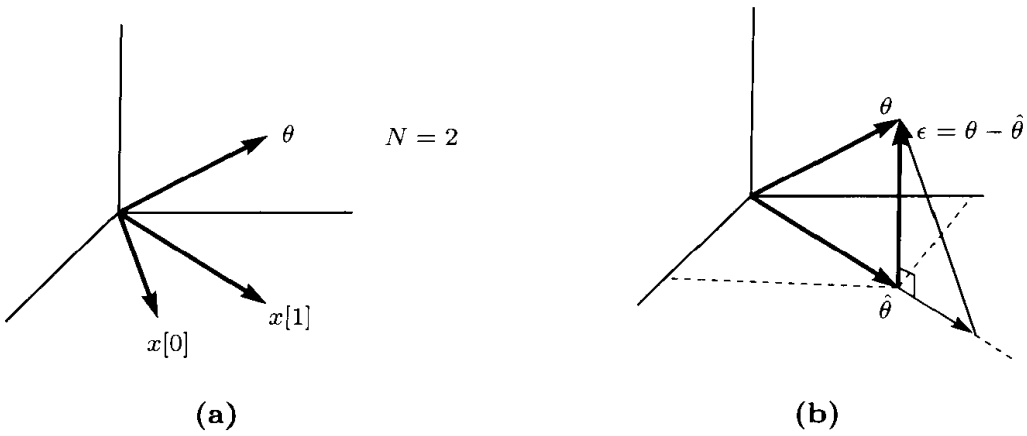


Figure 12.3 Orthogonality principle for LMMSE estimation

to minimize the MSE

$$\begin{aligned} E[(\theta - \hat{\theta})^2] &= E\left[\left(\theta - \sum_{n=0}^{N-1} a_n x[n]\right)^2\right] \\ &= \left\|\theta - \sum_{n=0}^{N-1} a_n x[n]\right\|^2. \end{aligned}$$

But this means that minimization of the MSE is equivalent to a minimization of the squared length of the *error vector* $\epsilon = \theta - \hat{\theta}$. The error vector is shown in Figure 12.3b for several candidate estimates. Clearly, the length of the error vector is minimized when ϵ is orthogonal to the subspace spanned by $\{x[0], x[1], \ldots, x[N-1]\}$. Hence, we require

$$\epsilon \perp x[0], x[1], \ldots, x[N-1] \tag{12.13}$$

or by using our definition of orthogonality

$$E\left[(\theta - \hat{\theta})x[n]\right] = 0 \qquad n = 0, 1, \ldots, N-1. \tag{12.14}$$

This is the important *orthogonality principle* or *projection theorem.* It says that *in estimating the realization of a random variable by a linear combination of data samples, the optimal estimator is obtained when the error is orthogonal to each data sample.* Using the orthogonality principle, the weighting coefficients are easily found as

$$E\left[\left(\theta - \sum_{n=0}^{N-1} a_m x[m]\right) x[n]\right] = 0 \qquad n = 0, 1, \ldots, N-1$$

or

$$\sum_{n=0}^{N-1} a_m E(x[m]x[n]) = E(\theta x[n]) \qquad n = 0, 1, \ldots, N-1.$$

In matrix form this is

$$\begin{bmatrix} E(x^2[0]) & E(x[0]x[1]) & \dots & E(x[0]x[N-1]) \\ E(x[1]x[0]) & E(x^2[1]) & \dots & E(x[1]x[N-1]) \\ \vdots & \vdots & \ddots & \vdots \\ E(x[N-1]x[0]) & E(x[N-1]x[1]) & \dots & E(x^2[N-1]) \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ \vdots \\ a_{N-1} \end{bmatrix}$$

$$= \begin{bmatrix} E(\theta x[0]) \\ E(\theta x[1]) \\ \vdots \\ E(\theta x[N-1]) \end{bmatrix}. \tag{12.15}$$

These are the *normal equations*. The matrix is recognized as $\mathbf{C}_{xx}$, and the right-hand vector as $\mathbf{C}_{x\theta}$. Therefore,

$$\mathbf{C}_{xx}\mathbf{a} = \mathbf{C}_{x\theta} \tag{12.16}$$

and

$$\mathbf{a} = \mathbf{C}_{xx}^{-1}\mathbf{C}_{x\theta}. \tag{12.17}$$

The LMMSE estimator of θ is

$$\begin{aligned} \hat{\theta} &= \mathbf{a}^T\mathbf{x} \\ &= \mathbf{C}_{x\theta}^T\mathbf{C}_{xx}^{-1}\mathbf{x} \end{aligned}$$

or finally

$$\hat{\theta} = \mathbf{C}_{\theta x}\mathbf{C}_{xx}^{-1}\mathbf{x} \tag{12.18}$$

in agreement with (12.7). The minimum Bayesian MSE is the squared length of the error vector or

$$\begin{aligned} \text{Bmse}(\hat{\theta}) &= ||\epsilon||^2 \\ &= \left\| \theta - \sum_{n=0}^{N-1} a_n x[n] \right\|^2 \\ &= E\left[\left(\theta - \sum_{n=0}^{N-1} a_n x[n] \right)^2 \right] \end{aligned}$$

for the a_n's given by (12.17). But

$$\begin{aligned} \text{Bmse}(\hat{\theta}) &= E\left[\left(\theta - \sum_{n=0}^{N-1} a_n x[n] \right) \left(\theta - \sum_{m=0}^{N-1} a_m x[m] \right) \right] \\ &= E\left[\left(\theta - \sum_{n=0}^{N-1} a_n x[n] \right) \theta \right] - E\left[\left(\theta - \sum_{n=0}^{N-1} a_n x[n] \right) \sum_{m=0}^{N-1} a_m x[m] \right] \\ &= E(\theta^2) - \sum_{n=0}^{N-1} a_n E(x[n]\theta) - \sum_{m=0}^{N-1} a_m E\left[\left(\theta - \sum_{n=0}^{N-1} a_n x[n] \right) x[m] \right]. \end{aligned}$$

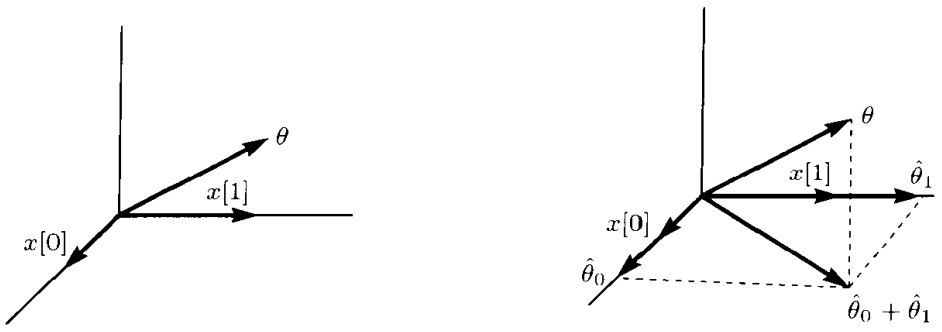


Figure 12.4 Linear estimation by orthogonal vectors

The last term is zero due to the orthogonality principle, producing

$$\begin{aligned} \text{Bmse}(\hat{\theta}) &= C_{\theta\theta} - \mathbf{a}^T \mathbf{C}_{x\theta} \\ &= C_{\theta\theta} - \mathbf{C}_{x\theta}^T \mathbf{C}_{xx}^{-1} \mathbf{C}_{x\theta} \\ &= C_{\theta\theta} - \mathbf{C}_{\theta x} \mathbf{C}_{xx}^{-1} \mathbf{C}_{x\theta} \end{aligned}$$

in agreement with (12.8).

Many important results can be easily derived when we view the LMMSE estimator in a vector space framework. In Section 12.6 we examine its application to sequential estimation. As a simple example of its utility in conceptualizing the estimation problem, we provide the following illustration.

Example 12.2 - Estimation by Orthogonal Vectors

Assume that $x[0]$ and $x[1]$ are zero mean and uncorrelated with each other. However, they are each correlated with θ. Figure 12.4a illustrates this case. The LMMSE estimator of θ based on $x[0]$ and $x[1]$ is the sum of the projections of θ on $x[0]$ and $x[1]$ as shown in Figure 12.4b or

$$\begin{aligned} \hat{\theta} &= \hat{\theta}_0 + \hat{\theta}_1 \\ &= \left(\theta, \frac{x[0]}{||x[0]||}\right) \frac{x[0]}{||x[0]||} + \left(\theta, \frac{x[1]}{||x[1]||}\right) \frac{x[1]}{||x[1]||}. \end{aligned}$$

Each component is the length of the projection $(\theta, x[n]/||x[n]||)$ times the unit vector in the $x[n]$ direction. Equivalently, since $||x[n]|| = \sqrt{\text{var}(x[n])}$ is a constant, this may be written

$$\hat{\theta} = \frac{(\theta, x[0])}{(x[0], x[0])} x[0] + \frac{(\theta, x[1])}{(x[1], x[1])} x[1].$$

By the definition of the inner product (12.10) we have

$$\hat{\theta} = \frac{E(\theta x[0])}{E(x^2[0])} x[0] + \frac{E(\theta x[1])}{E(x^2[1])} x[1]$$

$$= \begin{bmatrix} E(\theta x[0]) & E(\theta x[1]) \end{bmatrix} \begin{bmatrix} E(x^2[0]) & 0 \\ 0 & E(x^2[1]) \end{bmatrix}^{-1} \begin{bmatrix} x[0] \\ x[1] \end{bmatrix}$$

$$= \mathbf{C}_{\theta x}\mathbf{C}_{xx}^{-1}\mathbf{x}.$$

Clearly, the ease with which this result was obtained is due to the orthogonality of $x[0], x[1]$ or, equivalently, the diagonal nature of $\mathbf{C}_{xx}$. For nonorthogonal data samples the same approach can be used if we first orthogonalize the samples or replace the data by uncorrelated samples that span the same subspace. We will have more to say about this approach in Section 12.6. ◇

12.5 The Vector LMMSE Estimator

The vector LMMSE estimator is a straightforward extension of the scalar one. Now we wish to find the linear estimator that *minimizes the Bayesian MSE for each element.* We assume that

$$\hat{\theta}_i = \sum_{n=0}^{N-1} a_{in}x[n] + a_{iN} \tag{12.19}$$

for $i = 1, 2, \ldots, p$ and choose the weighting coefficients to minimize

$$\text{Bmse}(\hat{\theta}_i) = E\left[(\theta_i - \hat{\theta}_i)^2\right] \qquad i = 1, 2, \ldots, p$$

where the expectation is with respect to $p(\mathbf{x}, \theta_i)$. Since we are actually determining p separate estimators, the scalar solution can be applied, and we obtain from (12.6)

$$\hat{\theta}_i = E(\theta_i) + \mathbf{C}_{\theta_i x}\mathbf{C}_{xx}^{-1}(\mathbf{x} - E(\mathbf{x})) \qquad i = 1, 2, \ldots, p$$

and the minimum Bayesian MSE is from (12.8)

$$\text{Bmse}(\hat{\theta}_i) = C_{\theta_i\theta_i} - \mathbf{C}_{\theta_i x}\mathbf{C}_{xx}^{-1}\mathbf{C}_{x\theta_i} \qquad i = 1, 2, \ldots, p.$$

The scalar LMMSE estimators can be combined into a vector estimator as

$$\begin{aligned} \hat{\boldsymbol{\theta}} &= \begin{bmatrix} E(\theta_1) \\ E(\theta_2) \\ \vdots \\ E(\theta_p) \end{bmatrix} + \begin{bmatrix} \mathbf{C}_{\theta_1 x}\mathbf{C}_{xx}^{-1}(\mathbf{x} - E(\mathbf{x})) \\ \mathbf{C}_{\theta_2 x}\mathbf{C}_{xx}^{-1}(\mathbf{x} - E(\mathbf{x})) \\ \vdots \\ \mathbf{C}_{\theta_p x}\mathbf{C}_{xx}^{-1}(\mathbf{x} - E(\mathbf{x})) \end{bmatrix} \\ &= \begin{bmatrix} E(\theta_1) \\ E(\theta_2) \\ \vdots \\ E(\theta_p) \end{bmatrix} + \begin{bmatrix} \mathbf{C}_{\theta_1 x} \\ \mathbf{C}_{\theta_2 x} \\ \vdots \\ \mathbf{C}_{\theta_p x} \end{bmatrix} \mathbf{C}_{xx}^{-1}(\mathbf{x} - E(\mathbf{x})) \\ &= E(\boldsymbol{\theta}) + \mathbf{C}_{\theta x}\mathbf{C}_{xx}^{-1}(\mathbf{x} - E(\mathbf{x})) \end{aligned} \tag{12.20}$$

where now $\mathbf{C}_{\theta x}$ is a $p \times N$ matrix. By a similar approach we find that the Bayesian MSE matrix is (see Problem 12.7)

$$\begin{aligned} \mathbf{M}_{\hat{\boldsymbol{\theta}}} &= E\left[(\boldsymbol{\theta} - \hat{\boldsymbol{\theta}})(\boldsymbol{\theta} - \hat{\boldsymbol{\theta}})^T\right] \\ &= \mathbf{C}_{\theta\theta} - \mathbf{C}_{\theta x}\mathbf{C}_{xx}^{-1}\mathbf{C}_{x\theta} \end{aligned} \tag{12.21}$$

where $\mathbf{C}_{\theta\theta}$ is the $p \times p$ covariance matrix. Consequently, the minimum Bayesian MSE is (see Problem 12.7)

$$\text{Bmse}(\hat{\theta}_i) = [\mathbf{M}_{\hat{\theta}}]_{ii}. \tag{12.22}$$

Of course, these results are identical to those for the MMSE estimator in the Gaussian case, for which the estimator is linear. Note that to determine the LMMSE estimator we require only the first two moments of the PDF.

Two properties of the LMMSE estimator are particularly useful. The first one states that the LMMSE estimator commutes over linear (actually affine) transformations. This is to say that if

$$\boldsymbol{\alpha} = \mathbf{A}\boldsymbol{\theta} + \mathbf{b}$$

then the LMMSE estimator of $\boldsymbol{\alpha}$ is

$$\hat{\boldsymbol{\alpha}} = \mathbf{A}\hat{\boldsymbol{\theta}} + \mathbf{b} \tag{12.23}$$

with $\hat{\boldsymbol{\theta}}$ given by (12.20). The second property states that the LMMSE estimator of a sum of unknown parameters is the sum of the individual estimators. Specifically, if we wish to estimate $\boldsymbol{\alpha} = \boldsymbol{\theta}_1 + \boldsymbol{\theta}_2$, then

$$\hat{\boldsymbol{\alpha}} = \hat{\boldsymbol{\theta}}_1 + \hat{\boldsymbol{\theta}}_2 \tag{12.24}$$

where

$$\begin{aligned} \hat{\boldsymbol{\theta}}_1 &= E(\boldsymbol{\theta}_1) + \mathbf{C}_{\theta_1 x}\mathbf{C}_{xx}^{-1}(\mathbf{x} - E(\mathbf{x})) \\ \hat{\boldsymbol{\theta}}_2 &= E(\boldsymbol{\theta}_2) + \mathbf{C}_{\theta_2 x}\mathbf{C}_{xx}^{-1}(\mathbf{x} - E(\mathbf{x})). \end{aligned}$$

The proof of these properties is left to the reader as an exercise (see Problem 12.8).

In analogy with the BLUE there is a corresponding Gauss-Markov theorem for the Bayesian case. It asserts that for data having the Bayesian linear model *form*, i.e., the Bayesian linear model *without the Gaussian assumption*, an optimal linear estimator exists. Optimality is measured by the Bayesian MSE. The theorem is just the application of our LMMSE estimator to the Bayesian linear model. More specifically, the data are assumed to be

$$\mathbf{x} = \mathbf{H}\boldsymbol{\theta} + \mathbf{w}$$

where $\boldsymbol{\theta}$ is a random vector to be estimated and has mean $E(\boldsymbol{\theta})$ and covariance $\mathbf{C}_{\theta\theta}$, $\mathbf{H}$ is a known observation matrix, $\mathbf{w}$ is a random vector with zero mean and covariance $\mathbf{C}_w$,

and $\boldsymbol{\theta}$ and $\mathbf{w}$ are uncorrelated. Then, the LMMSE estimator of $\boldsymbol{\theta}$ is given by (12.20), where

$$\begin{aligned} E(\mathbf{x}) &= \mathbf{H}E(\boldsymbol{\theta}) \\ \mathbf{C}_{xx} &= \mathbf{H}\mathbf{C}_{\theta\theta}\mathbf{H}^T + \mathbf{C}_w \\ \mathbf{C}_{\theta x} &= \mathbf{C}_{\theta\theta}\mathbf{H}^T. \end{aligned}$$

(See Section 10.6 for details.) This is summarized by the Bayesian Gauss-Markov theorem.

Theorem 12.1 (Bayesian Gauss-Markov Theorem) *If the data are described by the Bayesian linear model form*

$$\mathbf{x} = \mathbf{H}\boldsymbol{\theta} + \mathbf{w} \tag{12.25}$$

where $\mathbf{x}$ is an $N \times 1$ data vector, $\mathbf{H}$ is a known $N \times p$ observation matrix, $\boldsymbol{\theta}$ is a $p \times 1$ random vector of parameters whose realization is to be estimated and has mean $E(\boldsymbol{\theta})$ and covariance matrix $\mathbf{C}_{\theta\theta}$, and $\mathbf{w}$ is an $N \times 1$ random vector with zero mean and covariance matrix $\mathbf{C}_w$ and is uncorrelated with $\boldsymbol{\theta}$ (the joint PDF $p(\mathbf{w}, \boldsymbol{\theta})$ is otherwise arbitrary), then the LMMSE estimator of $\boldsymbol{\theta}$ is

$$\begin{aligned} \hat{\boldsymbol{\theta}} &= E(\boldsymbol{\theta}) + \mathbf{C}_{\theta\theta}\mathbf{H}^T(\mathbf{H}\mathbf{C}_{\theta\theta}\mathbf{H}^T + \mathbf{C}_w)^{-1}(\mathbf{x} - \mathbf{H}E(\boldsymbol{\theta})) && (12.26) \\ &= E(\boldsymbol{\theta}) + (\mathbf{C}_{\theta\theta}^{-1} + \mathbf{H}^T\mathbf{C}_w^{-1}\mathbf{H})^{-1}\mathbf{H}^T\mathbf{C}_w^{-1}(\mathbf{x} - \mathbf{H}E(\boldsymbol{\theta})). && (12.27) \end{aligned}$$

The performance of the estimator is measured by the error $\boldsymbol{\epsilon} = \boldsymbol{\theta} - \hat{\boldsymbol{\theta}}$ whose mean is zero and whose covariance matrix is

$$\begin{aligned} \mathbf{C}_\epsilon &= E_{x,\theta}(\boldsymbol{\epsilon}\boldsymbol{\epsilon}^T) \\ &= \mathbf{C}_{\theta\theta} - \mathbf{C}_{\theta\theta}\mathbf{H}^T(\mathbf{H}\mathbf{C}_{\theta\theta}\mathbf{H}^T + \mathbf{C}_w)^{-1}\mathbf{H}\mathbf{C}_{\theta\theta} && (12.28) \\ &= (\mathbf{C}_{\theta\theta}^{-1} + \mathbf{H}^T\mathbf{C}_w^{-1}\mathbf{H})^{-1}. && (12.29) \end{aligned}$$

The error covariance matrix is also the minimum MSE matrix $\mathbf{M}_{\hat{\theta}}$ whose diagonal elements yield the minimum Bayesian MSE

$$\begin{aligned} [\mathbf{M}_{\hat{\theta}}]_{ii} &= [\mathbf{C}_\epsilon]_{ii} \\ &= \text{Bmse}(\hat{\theta}_i). && (12.30) \end{aligned}$$

These results are identical to those in Theorem 11.1 for the Bayesian linear model except that the error vector is not necessarily Gaussian. An example of the determination of this estimator and its minimum Bayesian MSE has already been given in Section 10.6. The Bayesian Gauss-Markov theorem states that within the class of linear estimators the one that minimizes the Bayesian MSE for each element of $\boldsymbol{\theta}$ is given by (12.26) or (12.27). It will not be optimal unless the conditional expectation $E(\boldsymbol{\theta}|\mathbf{x})$ happens to be linear. Such was the case for the jointly Gaussian PDF. Although suboptimal, the LMMSE estimator is in practice quite useful, being available in closed form and depending only on the means and covariances.

12.6 Sequential LMMSE Estimation

In Chapter 8 we discussed the sequential LS procedure as the process of updating in time the LSE estimator as new data become available. An analogous procedure can be used for LMMSE estimators. That this is possible follows from the vector space viewpoint. In Example 12.2 the LMMSE estimator was obtained by adding to the old estimate $\hat{\theta}_0$, that based on $x[0]$, the estimate $\hat{\theta}_1$, that based on the new datum $x[1]$. This was possible because $x[0]$ and $x[1]$ were orthogonal. When they are not, the algebra becomes more tedious, although the approach is similar. Before stating the general results we will illustrate the basic operations involved by considering the DC level in white noise, which has the Bayesian linear model form. The derivation will be purely algebraic. Then we will repeat the derivation but appeal to the vector space approach. The reason for doing so will be to "lay the groundwork" for the Kalman filter in Chapter 13, which will be derived using the same approach. We will assume a zero mean for the DC level A, so that the vector space approach is applicable.

To begin the algebraic derivation we have from Example 10.1 with $\mu_A = 0$ (see (10.11))

$$\hat{A}[N-1] = \frac{\sigma_A^2}{\sigma_A^2 + \frac{\sigma^2}{N}} \bar{x}$$

where $\hat{A}[N-1]$ denotes the LMMSE estimator based on $\{x[0], x[1], \ldots, x[N-1]\}$. This is because the LMMSE estimator is identical in form to the MMSE estimator for the Gaussian case. Also, we note that from (10.14)

$$\text{Bmse}(\hat{A}[N-1]) = \frac{\sigma_A^2 \sigma^2}{N\sigma_A^2 + \sigma^2}. \tag{12.31}$$

To update our estimator as $x[N]$ becomes available

$$\begin{aligned}
\hat{A}[N] &= \frac{\sigma_A^2}{\sigma_A^2 + \frac{\sigma^2}{N+1}} \frac{1}{N+1} \sum_{n=0}^{N} x[n] \\
&= \frac{N\sigma_A^2}{(N+1)\sigma_A^2 + \sigma^2} \frac{1}{N} \left(\sum_{n=0}^{N-1} x[n] + x[N] \right) \\
&= \frac{N\sigma_A^2}{(N+1)\sigma_A^2 + \sigma^2} \frac{\sigma_A^2 + \frac{\sigma^2}{N}}{\sigma_A^2} \hat{A}[N-1] + \frac{\sigma_A^2}{(N+1)\sigma_A^2 + \sigma^2} x[N] \\
&= \frac{N\sigma_A^2 + \sigma^2}{(N+1)\sigma_A^2 + \sigma^2} \hat{A}[N-1] + \frac{\sigma_A^2}{(N+1)\sigma_A^2 + \sigma^2} x[N] \\
&= \hat{A}[N-1] + \left(\frac{N\sigma_A^2 + \sigma^2}{(N+1)\sigma_A^2 + \sigma^2} - 1 \right) \hat{A}[N-1] + \frac{\sigma_A^2}{(N+1)\sigma_A^2 + \sigma^2} x[N] \\
&= \hat{A}[N-1] + \frac{\sigma_A^2}{(N+1)\sigma_A^2 + \sigma^2} (x[N] - \hat{A}[N-1]).
\end{aligned} \tag{12.32}$$

Similar to the sequential LS estimator we correct the old estimate $\hat{A}[N-1]$ by a scaled version of the prediction error $x[N] - \hat{A}[N-1]$. The scaling or gain factor is from (12.32) and (12.31)

$$\begin{aligned} K[N] &= \frac{\sigma_A^2}{(N+1)\sigma_A^2 + \sigma^2} \\ &= \frac{\text{Bmse}(\hat{A}[N-1])}{\text{Bmse}(\hat{A}[N-1]) + \sigma^2} \end{aligned} \tag{12.33}$$

and decreases to zero as $N \to \infty$, reflecting the increased confidence in the old estimator. We can also update the minimum Bayesian MSE since from (12.31)

$$\begin{aligned} \text{Bmse}(\hat{A}[N]) &= \frac{\sigma_A^2 \sigma^2}{(N+1)\sigma_A^2 + \sigma^2} \\ &= \frac{N\sigma_A^2 + \sigma^2}{(N+1)\sigma_A^2 + \sigma^2} \frac{\sigma_A^2 \sigma^2}{N\sigma_A^2 + \sigma^2} \\ &= (1 - K[N])\text{Bmse}(\hat{A}[N-1]). \end{aligned}$$

Summarizing our results we have the following sequential LMMSE estimator.

Estimator Update:

$$\hat{A}[N] = \hat{A}[N-1] + K[N](x[N] - \hat{A}[N-1]) \tag{12.34}$$

where

$$K[N] = \frac{\text{Bmse}(\hat{A}[N-1])}{\text{Bmse}(\hat{A}[N-1]) + \sigma^2}. \tag{12.35}$$

Minimum MSE Update:

$$\text{Bmse}(\hat{A}[N]) = (1 - K[N])\text{Bmse}(\hat{A}[N-1]). \tag{12.36}$$

We now use the vector space viewpoint to derive the same results. (The general sequential LMMSE estimator is derived in Appendix 12A using the vector space approach. It is a straightforward extension of the following derivation.) Assume that the LMMSE estimator $\hat{A}[1]$ is to be found, which is based on the data $\{x[0], x[1]\}$ as shown in Figure 12.5a. Because $x[0]$ and $x[1]$ are not orthogonal, we cannot simply add the estimate based on $x[0]$ to the estimate based on $x[1]$. If we did, we would be adding in an extra component along $x[0]$. However, we can form $\hat{A}[1]$ as the sum of $\hat{A}[0]$ and a component *orthogonal* to $\hat{A}[0]$ as shown in Figure 12.5b. That component is $\Delta\hat{A}[1]$. To find a vector in the direction of this component recall that the LMMSE estimator has the property that the error is orthogonal to the data. Hence, if we find the LMMSE estimator of $x[1]$ based on $x[0]$, call it $\hat{x}[1|0]$, the error $x[1] - \hat{x}[1|0]$ will be orthogonal to $x[0]$. This

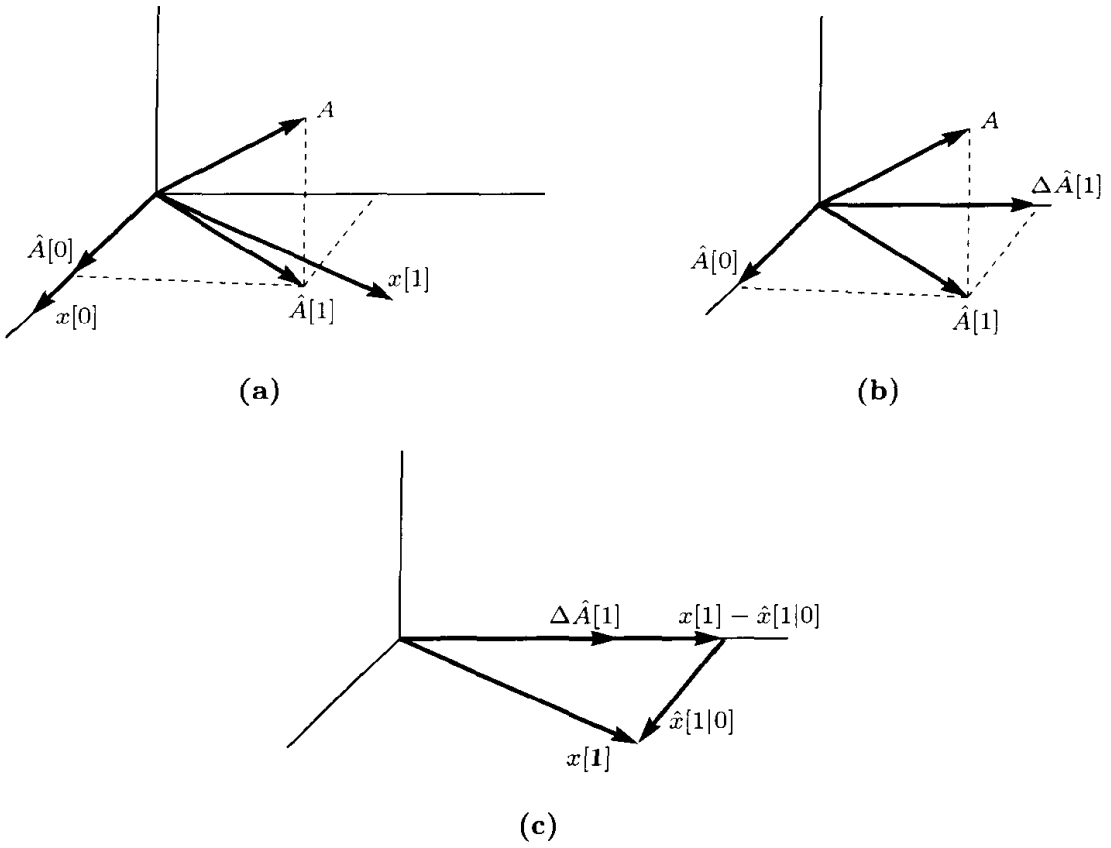


Figure 12.5 Sequential estimation using vector space approach

is shown in Figure 12.5c. Since $x[0]$ and $x[1] - \hat{x}[1|0]$ are orthogonal, we can project A onto each vector separately and add the results, so that

$$\hat{A}[1] = \hat{A}[0] + \Delta\hat{A}[1].$$

To find the correction term $\Delta\hat{A}[1]$ first note that the LMMSE estimator of a random variable y based on x, where both are zero mean, is from (12.7)

$$\hat{y} = \frac{E(xy)}{E(x^2)}x. \tag{12.37}$$

Thus, we have that

$$\begin{aligned} \hat{x}[1|0] &= \frac{E(x[0]x[1])}{E(x^2[0])}x[0] \\ &= \frac{E\left[(A + w[0])(A + w[1])\right]}{E\left[(A + w[0])^2\right]}x[0] \\ &= \frac{\sigma_A^2}{\sigma_A^2 + \sigma^2}x[0] \end{aligned} \tag{12.38}$$

and the error vector $\tilde{x}[1] = x[1] - \hat{x}[1|0]$ represents the new information that $x[1]$ contributes to the estimation of A. As such, it is called the *innovation*. The projection of A along this error vector is the desired correction

$$\begin{aligned}\Delta\hat{A}[1] &= \left(A, \frac{\tilde{x}[1]}{||\tilde{x}[1]||}\right)\frac{\tilde{x}[1]}{||\tilde{x}[1]||} \\ &= \frac{E(A\tilde{x}[1])\tilde{x}[1]}{E(\tilde{x}^2[1])}.\end{aligned}$$

If we let $K[1] = E(A\tilde{x}[1])/E(\tilde{x}^2[1])$, we have for the LMMSE estimator

$$\hat{A}[1] = \hat{A}[0] + K[1](x[1] - \hat{x}[1|0]).$$

To evaluate $\hat{x}[1|0]$ we note that $x[1] = A + w[1]$. Hence, from the additive property (12.24) $\hat{x}[1|0] = \hat{A}[0] + \hat{w}[1|0]$. Since $w[1]$ is uncorrelated with $w[0]$, we have $\hat{w}[1|0] = 0$. Finally, then $\hat{x}[1|0] = \hat{A}[0]$, and $\hat{A}[0]$ is found as

$$\hat{A}[0] = \frac{E(Ax[0])}{E(x^2[0])}x[0] = \frac{\sigma_A^2}{\sigma_A^2 + \sigma^2}x[0].$$

Thus,

$$\hat{A}[1] = \hat{A}[0] + K[1](x[1] - \hat{A}[0]).$$

It remains only to determine the gain $K[1]$ for the correction. Since

$$\begin{aligned}\tilde{x}[1] &= x[1] - \hat{x}[1|0] \\ &= x[1] - \hat{A}[0] \\ &= x[1] - \frac{\sigma_A^2}{\sigma_A^2 + \sigma^2}x[0]\end{aligned}$$

the gain becomes

$$\begin{aligned}K[1] &= \frac{E\left[A\left(x[1] - \frac{\sigma_A^2}{\sigma_A^2 + \sigma^2}x[0]\right)\right]}{E\left[\left(x[1] - \frac{\sigma_A^2}{\sigma_A^2 + \sigma^2}x[0]\right)^2\right]} \\ &= \frac{\sigma_A^2}{2\sigma_A^2 + \sigma^2}\end{aligned}$$

which agrees with (12.33) for $N = 1$. We can continue this procedure to find $\hat{A}[2]$, $\hat{A}[3]$, ... In general, we

1. find the LMMSE estimator of A based on $x[0]$, yielding $\hat{A}[0]$
2. find the LMMSE estimator of $x[1]$ based on $x[0]$, yielding $\hat{x}[1|0]$

3. determine the innovation of the new datum $x[1]$, yielding $x[1] - \hat{x}[1|0]$

4. add to $\hat{A}[0]$ the LMMSE estimator of A based on the innovation, yielding $\hat{A}[1]$

5. continue the process.

In essence we are generating a set of *uncorrelated or orthogonal* random variables, namely, the innovations $\{x[0], x[1]-\hat{x}[1|0], x[2]-\hat{x}[2|0,1], \ldots\}$. This procedure is termed a Gram-Schmidt orthogonalization (see Problem 12.10). To find the LMMSE estimator of A based on $\{x[0], x[1], \ldots, x[N-1]\}$ we simply *add* the individual estimators to yield

$$\hat{A}[N-1] = \sum_{n=0}^{N-1} K[n]\left(x[n] - \hat{x}[n|0,1,\ldots,n-1]\right) \tag{12.39}$$

where each gain factor is

$$K[n] = \frac{E\left[A\left(x[n] - \hat{x}[n|0,1,\ldots,n-1]\right)\right]}{E\left[\left(x[n] - \hat{x}[n|0,1,\ldots,n-1]\right)^2\right]}. \tag{12.40}$$

This simple form is due to the uncorrelated property of the innovations. In sequential form (12.39) becomes

$$\hat{A}[N] = \hat{A}[N-1] + K[N](x[N] - \hat{x}[N|0,1,\ldots,N-1]). \tag{12.41}$$

To complete the derivation we must find $\hat{x}[N|0,1,\ldots,N-1]$ and the gain $K[N]$. We first use the additive property (12.24). Since A and $w[N]$ are uncorrelated and zero mean, the LMMSE estimator of $x[N] = A + w[N]$ is

$$\hat{x}[N|0,1,\ldots,N-1] = \hat{A}[N|0,1,\ldots,N-1] + \hat{w}[N|0,1,\ldots,N-1].$$

But $\hat{A}[N|0,1,\ldots,N-1]$ is by definition the LMMSE estimator of the realization of A at time N based on $\{x[0], x[1], \ldots, x[N-1]\}$ or $\hat{A}[N-1]$ (recalling that A does not change over the observation interval). Also, $\hat{w}[N|0,1,\ldots,N-1]$ is zero since $w[N]$ is uncorrelated with the previous data samples since A and $w[n]$ are uncorrelated and $w[n]$ is white noise. Thus,

$$\hat{x}[N|0,1,\ldots,N-1] = \hat{A}[N-1]. \tag{12.42}$$

To find the gain we use (12.40) and (12.42):

$$K[N] = \frac{E\left[A(x[N] - \hat{A}[N-1])\right]}{E\left[(x[N] - \hat{A}[N-1])^2\right]}.$$

But

$$E\left[A(x[N] - \hat{A}[N-1])\right] = E\left[(A - \hat{A}[N-1])(x[N] - \hat{A}[N-1])\right] \tag{12.43}$$

since $x[N] - \hat{A}[N-1]$ is the innovation of $x[N]$ which is orthogonal to $\{x[0], x[1], \ldots, x[N-1]\}$ and hence to $\hat{A}[N-1]$ (a linear combination of these data samples). Also, $E[w[N](A - \hat{A}[N-1])] = 0$ as explained previously, so that

$$\begin{aligned} E\left[A(x[N] - \hat{A}[N-1])\right] &= E\left[(A - \hat{A}[N-1])^2\right] \\ &= \text{Bmse}(\hat{A}[N-1]). \end{aligned} \tag{12.44}$$

Also,

$$\begin{aligned} E\left[(x[N] - \hat{A}[N-1])^2\right] &= E\left[(w[N] + A - \hat{A}[N-1])^2\right] \\ &= E(w^2[N]) + E\left[(A - \hat{A}[N-1])^2\right] \\ &= \sigma^2 + \text{Bmse}(\hat{A}[N-1]) \end{aligned} \tag{12.45}$$

so that finally

$$K[N] = \frac{\text{Bmse}(\hat{A}[N-1])}{\sigma^2 + \text{Bmse}(\hat{A}[N-1])}. \tag{12.46}$$

The minimum MSE update can be obtained as

$$\begin{aligned} \text{Bmse}(\hat{A}[N]) &= E\left[(A - \hat{A}[N])^2\right] \\ &= E\left[\left(A - \hat{A}[N-1] - K[N](x[N] - \hat{A}[N-1])\right)^2\right] \\ &= E\left[(A - \hat{A}[N-1])^2\right] - 2K[N]E\left[(A - \hat{A}[N-1])(x[N] - \hat{A}[N-1])\right] \\ &\quad + K^2[N]E\left[(x[N] - \hat{A}[N-1])^2\right] \\ &= \text{Bmse}(\hat{A}[N-1]) - 2K[N]\text{Bmse}(\hat{A}[N-1]) \\ &\quad + K^2[N]\left(\sigma^2 + \text{Bmse}(\hat{A}[N-1])\right) \end{aligned}$$

where we have used (12.43)–(12.45). Using (12.46), we have

$$\text{Bmse}(\hat{A}[N]) = (1 - K[N])\text{Bmse}(\hat{A}[N-1]).$$

We have now derived the sequential LMMSE estimator equations based on the vector space viewpoint.

In Appendix 12A we generalize the preceding vector space derivation for the sequential vector LMMSE estimator. To do so we must assume the data has the Bayesian linear model form (see Theorem 12.1) and that $\mathbf{C}_w$ is a *diagonal* matrix, so that the $w[n]$*'s are uncorrelated* with variance $E(w^2[n]) = \sigma_n^2$. The latter is a critical assumption and was used in the previous example. Fortuitously, the set of equations obtained are valid for the case of nonzero means as well (see Appendix 12A). Hence, the equations to follow are the sequential implementation of the vector LMMSE estimator of (12.26)

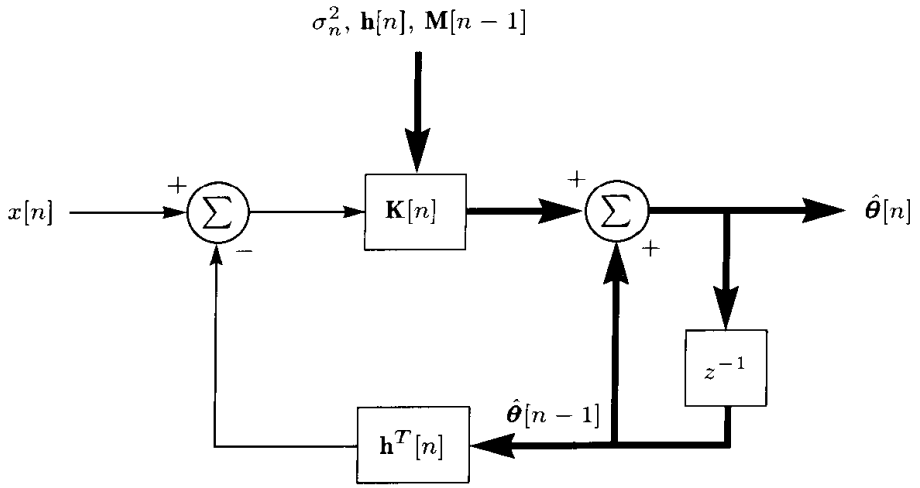


Figure 12.6 Sequential linear minimum mean square estimator

or (12.27) when the noise covariance matrix $\mathbf{C}_w$ is diagonal. To define the equations we let $\hat{\boldsymbol{\theta}}[n]$ be the LMMSE estimator based on $\{x[0], x[1], \ldots, x[n]\}$, and $\mathbf{M}[n]$ be the corresponding minimum MSE matrix (just the sequential version of (12.28) or (12.29)) or

$$\mathbf{M}[n] = E\left[(\boldsymbol{\theta} - \hat{\boldsymbol{\theta}}[n])(\boldsymbol{\theta} - \hat{\boldsymbol{\theta}}[n])^T\right].$$

Also, we partition the $(n+1) \times p$ observation matrix as

$$\mathbf{H}[n] = \begin{bmatrix} \mathbf{H}[n-1] \\ \mathbf{h}^T[n] \end{bmatrix} = \begin{bmatrix} n \times p \\ 1 \times p \end{bmatrix}.$$

Then, the sequential LMMSE estimator becomes (see Appendix 12A)

Estimator Update:

$$\hat{\boldsymbol{\theta}}[n] = \hat{\boldsymbol{\theta}}[n-1] + \mathbf{K}[n](x[n] - \mathbf{h}^T[n]\hat{\boldsymbol{\theta}}[n-1]) \tag{12.47}$$

where

$$\mathbf{K}[n] = \frac{\mathbf{M}[n-1]\mathbf{h}[n]}{\sigma_n^2 + \mathbf{h}^T[n]\mathbf{M}[n-1]\mathbf{h}[n]}. \tag{12.48}$$

Minimum MSE Matrix Update:

$$\mathbf{M}[n] = (\mathbf{I} - \mathbf{K}[n]\mathbf{h}^T[n])\mathbf{M}[n-1]. \tag{12.49}$$

The gain factor $\mathbf{K}[n]$ is a $p \times 1$ vector, and the minimum MSE matrix has dimension $p \times p$. The entire estimator is summarized in Figure 12.6, where the thick arrows indicate vector processing. To start the recursion we need to specify initial values for $\hat{\boldsymbol{\theta}}[n-1]$ and $\mathbf{M}[n-1]$, so that $\mathbf{K}[n]$ can be determined from (12.48) and then $\hat{\boldsymbol{\theta}}[n]$ from (12.47). To do so we specify $\hat{\boldsymbol{\theta}}[-1]$ and $\mathbf{M}[-1]$ so that the recursion can begin at $n = 0$.

Since no data have been observed at $n = -1$, then from (12.19) our estimator becomes the constant $\hat{\theta}_i = a_{i0}$. It is easily shown that the LMMSE estimator is just the mean of θ_i (see also (12.3)) or

$$\hat{\boldsymbol{\theta}}[-1] = E(\boldsymbol{\theta}).$$

As a result, the minimum MSE matrix is

$$\begin{aligned} \mathbf{M}[-1] &= E\left[(\boldsymbol{\theta} - \hat{\boldsymbol{\theta}}[-1])(\boldsymbol{\theta} - \hat{\boldsymbol{\theta}}[-1])^T\right] \\ &= \mathbf{C}_{\theta\theta}. \end{aligned}$$

Several observations are in order.

1. For no prior knowledge about $\boldsymbol{\theta}$ we can let $\mathbf{C}_{\theta\theta} \to \infty$. Then, we have the same *form* as the sequential LSE (see Section 8.7), although the approaches are fundamentally different.

2. No matrix inversions are required.

3. The gain factor $\mathbf{K}[n]$ depends on our confidence in the new data sample as measured by σ_n^2 versus that in the previous data as summarized by $\mathbf{M}[n-1]$.

An example follows.

Example 12.3 - Bayesian Fourier Analysis

We now use sequential LS to compute the LMMSE estimator in Example 11.1. The data model is

$$x[n] = a\cos 2\pi f_0 n + b\sin 2\pi f_0 n + w[n] \qquad n = 0, 1, \ldots, N-1$$

where f_0 is any frequency (in Example 11.1 we assumed f_0 to be a multiple of $1/N$), excepting 0 and $1/2$ for which $\sin 2\pi f_0 n$ is identically zero, and $w[n]$ is white noise with variance σ^2. It is desired to estimate $\boldsymbol{\theta} = [a\, b]^T$ sequentially. We further assume that the Bayesian linear model form applies and that $\boldsymbol{\theta}$ has mean $E(\boldsymbol{\theta})$ and covariance $\sigma_\theta^2\mathbf{I}$. The sequential estimator is initialized by

$$\begin{aligned} \hat{\boldsymbol{\theta}}[-1] &= E(\boldsymbol{\theta}) \\ \mathbf{M}[-1] &= \mathbf{C}_{\theta\theta} = \sigma_\theta^2\mathbf{I}. \end{aligned}$$

Then, from (12.47)

$$\hat{\boldsymbol{\theta}}[0] = \hat{\boldsymbol{\theta}}[-1] + \mathbf{K}[0](x[0] - \mathbf{h}^T[0]\hat{\boldsymbol{\theta}}[-1]).$$

The $\mathbf{h}^T[0]$ vector is the first row of the observation matrix or

$$\mathbf{h}^T[0] = \begin{bmatrix} 1 & 0 \end{bmatrix}.$$

Subsequent rows are

$$\mathbf{h}^T[n] = \begin{bmatrix} \cos 2\pi f_0 n & \sin 2\pi f_0 n \end{bmatrix}.$$

The 2×1 gain vector is, from (12.48),

$$\mathbf{K}[0] = \frac{\mathbf{M}[-1]\mathbf{h}[0]}{\sigma^2 + \mathbf{h}^T[0]\mathbf{M}[-1]\mathbf{h}[0]}$$

where $\mathbf{M}[-1] = \sigma_\theta^2\mathbf{I}$. Once the gain vector has been found, $\hat{\boldsymbol{\theta}}[0]$ can be computed. Finally, the MSE matrix update is, from (12.49),

$$\mathbf{M}[0] = (\mathbf{I} - \mathbf{K}[0]\mathbf{h}^T[0])\mathbf{M}[-1]$$

and the procedure continues in like manner for $n \geq 1$. ◇

12.7 Signal Processing Examples - Wiener Filtering

We now examine in detail some of the important applications of the LMMSE estimator. In doing so we will assume that the data $\{x[0], x[1], \ldots, x[N-1]\}$ is WSS with zero mean. As such, the $N \times N$ covariance matrix $\mathbf{C}_{xx}$ takes the symmetric Toeplitz form

$$\begin{aligned} \mathbf{C}_{xx} &= \begin{bmatrix} r_{xx}[0] & r_{xx}[1] & \cdots & r_{xx}[N-1] \\ r_{xx}[1] & r_{xx}[0] & \cdots & r_{xx}[N-2] \\ \vdots & \vdots & \ddots & \vdots \\ r_{xx}[N-1] & r_{xx}[N-2] & \cdots & r_{xx}[0] \end{bmatrix} \\ &= \mathbf{R}_{xx} \end{aligned} \tag{12.50}$$

where $r_{xx}[k]$ is the ACF of the $x[n]$ process and $\mathbf{R}_{xx}$ denotes the autocorrelation matrix. Furthermore, the parameter $\boldsymbol{\theta}$ to be estimated is also assumed to be zero mean. The group of applications that we will describe are generically called *Wiener filters.*

There are three main problems that we will study. They are (see Figure 12.7)

1. *Filtering*, where $\theta = s[n]$ is to be estimated based on $x[m] = s[m] + w[m]$ for $m = 0, 1, \ldots, n$. The sequences $s[n]$ and $w[n]$ represent signal and noise processes. The problem is to filter the signal from the noise. Note that the signal sample is estimated based on the *present and past data only*, so that as n increases, we view the estimation process as the application of a causal filter to the data.

2. *Smoothing*, where $\theta = s[n]$ is to be estimated for $n = 0, 1, \ldots, N-1$ based on the data set $\{x[0], x[1], \ldots, x[N-1]\}$, where $x[n] = s[n] + w[n]$. In contrast to filtering we are allowed to use future data. For instance, to estimate $s[1]$ we can use the entire data set $\{x[0], x[1], \ldots, x[N-1]\}$, while in filtering we are constrained to use only $\{x[0], x[1]\}$. Clearly, in smoothing an estimate cannot be obtained until all the data has been collected.

3. *Prediction*, where $\theta = x[N-1+l]$ for l a positive integer is to be estimated based on $\{x[0], x[1], \ldots, x[N-1]\}$. This is referred to as the l-step prediction problem.

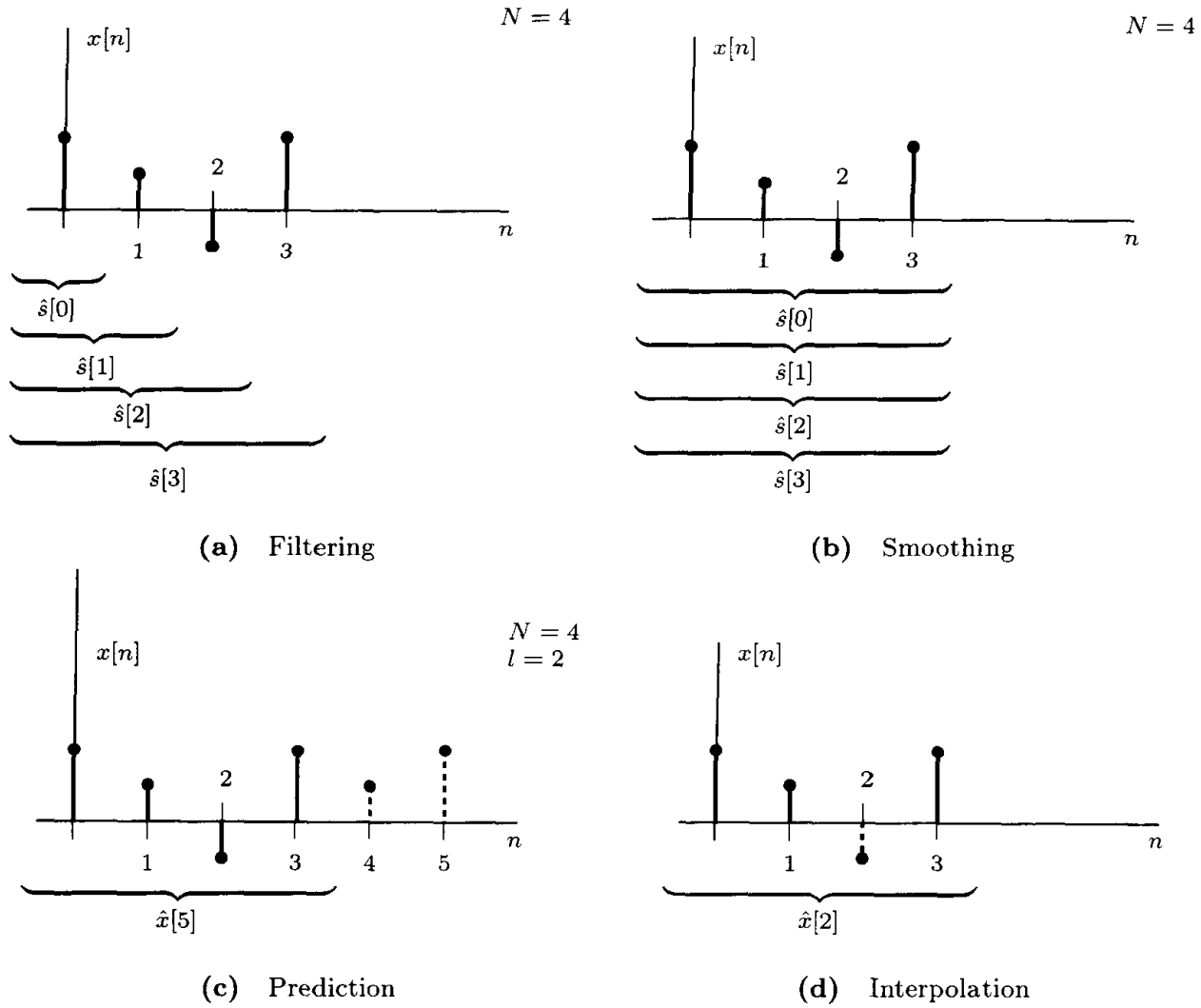


Figure 12.7 Wiener filtering problem definitions

A related problem that is explored in Problem 12.14 is to estimate $x[n]$ based on $\{x[0], \ldots, x[n-1], x[n+1], \ldots, x[N-1]\}$ and is termed the *interpolation* problem. To solve all three problems we use (12.20) with $E(\boldsymbol{\theta}) = E(\mathbf{x}) = \mathbf{0}$ or

$$\hat{\boldsymbol{\theta}} = \mathbf{C}_{\theta x}\mathbf{C}_{xx}^{-1}\mathbf{x} \tag{12.51}$$

and the minimum Bayesian MSE matrix given by (12.21)

$$\mathbf{M}_{\hat{\theta}} = \mathbf{C}_{\theta\theta} - \mathbf{C}_{\theta x}\mathbf{C}_{xx}^{-1}\mathbf{C}_{x\theta}. \tag{12.52}$$

Consider first the smoothing problem. We wish to estimate $\boldsymbol{\theta} = \mathbf{s} = [s[0]\, s[1] \ldots s[N-1]]^T$ based on $\mathbf{x} = [x[0]\, x[1] \ldots x[N-1]]^T$. We will make the reasonable assumption that the signal and noise processes are uncorrelated. Hence,

$$r_{xx}[k] = r_{ss}[k] + r_{ww}[k].$$

Then, we have

$$\mathbf{C}_{xx} = \mathbf{R}_{xx} = \mathbf{R}_{ss} + \mathbf{R}_{ww}.$$

Also,

$$\mathbf{C}_{\theta x} = E(\mathbf{s}\mathbf{x}^T) = E(\mathbf{s}(\mathbf{s}+\mathbf{w})^T) = \mathbf{R}_{ss}.$$

Therefore, the Wiener estimator of the signal is, from (12.51),

$$\hat{\mathbf{s}} = \mathbf{R}_{ss}(\mathbf{R}_{ss} + \mathbf{R}_{ww})^{-1}\mathbf{x}. \tag{12.53}$$

The $N \times N$ matrix

$$\mathbf{W} = \mathbf{R}_{ss}(\mathbf{R}_{ss} + \mathbf{R}_{ww})^{-1} \tag{12.54}$$

is referred to as the *Wiener smoothing matrix*. The corresponding minimum MSE matrix is, from (12.52),

$$\begin{aligned} \mathbf{M}_{\hat{s}} &= \mathbf{R}_{ss} - \mathbf{R}_{ss}(\mathbf{R}_{ss} + \mathbf{R}_{ww})^{-1}\mathbf{R}_{ss} \\ &= (\mathbf{I} - \mathbf{W})\mathbf{R}_{ss}. \end{aligned} \tag{12.55}$$

As an example, if $N = 1$ we would wish to estimate $s[0]$ based on $x[0] = s[0] + w[0]$. Then, the Wiener smoother $\mathbf{W}$ is just a scalar W given by

$$W = \frac{r_{ss}[0]}{r_{ss}[0] + r_{ww}[0]} = \frac{\eta}{\eta + 1} \tag{12.56}$$

where $\eta = r_{ss}[0]/r_{ww}[0]$ is the SNR. For a high SNR so that $W \to 1$, we have $\hat{s}[0] \to x[0]$, while for a low SNR so that $W \to 0$, we have $\hat{s}[0] \to 0$. The corresponding minimum MSE is

$$M_{\hat{s}} = (1 - W)r_{ss}[0] = \left(1 - \frac{\eta}{\eta + 1}\right) r_{ss}[0]$$

which for these two extremes is either 0 for a high SNR or $r_{ss}[0]$ for a low SNR. A numerical example of a Wiener smoother has been given in Section 11.7. See also Problem 12.15.

We next consider the filtering problem in which we wish to estimate $\theta = s[n]$ based on $\mathbf{x} = [x[0]\, x[1] \ldots x[n]]^T$. The problem is repeated for each value of n until the entire signal $s[n]$ for $n = 0, 1, \ldots, N-1$ has been estimated. As before, $x[n] = s[n] + w[n]$, where $s[n]$ and $w[n]$ are signal and noise processes that are uncorrelated with each other. Thus,

$$\mathbf{C}_{xx} = \mathbf{R}_{ss} + \mathbf{R}_{ww}$$

where $\mathbf{R}_{ss}, \mathbf{R}_{ww}$ are $(n+1) \times (n+1)$ autocorrelation matrices. Also,

$$\begin{aligned} \mathbf{C}_{\theta x} &= E\left(s[n]\left[\begin{array}{cccc} x[0] & x[1] & \ldots & x[n] \end{array}\right]\right) \\ &= E\left(s[n]\left[\begin{array}{cccc} s[0] & s[1] & \ldots & s[n] \end{array}\right]\right) \\ &= \left[r_{ss}[n]\, r_{ss}[n-1] \ldots r_{ss}[0]\right]. \end{aligned}$$

Letting the latter row vector be denoted as $\mathbf{r}_{ss}'^T$, we have from (12.51)

$$\hat{s}[n] = \mathbf{r}_{ss}'^T \left(\mathbf{R}_{ss} + \mathbf{R}_{ww}\right)^{-1} \mathbf{x}. \tag{12.57}$$

The $(n+1) \times 1$ vector of weights is seen to be

$$\mathbf{a} = (\mathbf{R}_{ss} + \mathbf{R}_{ww})^{-1} \mathbf{r}'_{ss}$$

recalling that

$$\hat{s}[n] = \mathbf{a}^T \mathbf{x}$$

where $\mathbf{a} = [a_0\, a_1 \ldots a_n]^T$ as in our original formulation of the scalar LMMSE estimator of (12.1). We may interpret the process of forming the estimator as n increases as a filtering operation if we define a *time varying* impulse response $h^{(n)}[k]$. Specifically, we let $h^{(n)}[k]$ be the response of a filter at time n to an impulse applied k samples before. To make the filtering correspondence we let

$$h^{(n)}[k] = a_{n-k} \qquad k = 0, 1, \ldots, n.$$

Note for future reference that the vector $\mathbf{h} = [h^{(n)}[0]\, h^{(n)}[1] \ldots h^{(n)}[n]]^T$ is just the vector $\mathbf{a}$ when flipped upside down. Then,

$$\begin{aligned} \hat{s}[n] &= \sum_{k=0}^{n} a_k x[k] \\ &= \sum_{k=0}^{n} h^{(n)}[n-k] x[k] \end{aligned}$$

or

$$\hat{s}[n] = \sum_{k=0}^{n} h^{(n)}[k] x[n-k] \tag{12.58}$$

which is recognized as a time varying FIR filter. To explicitly find the impulse response $\mathbf{h}$ we note that since

$$(\mathbf{R}_{ss} + \mathbf{R}_{ww})\, \mathbf{a} = \mathbf{r}'_{ss}$$

it follows that

$$(\mathbf{R}_{ss} + \mathbf{R}_{ww})\, \mathbf{h} = \mathbf{r}_{ss}$$

where $\mathbf{r}_{ss} = [r_{ss}[0]\, r_{ss}[1] \ldots r_{ss}[n]]^T$. This result depends on the symmetric Toeplitz nature of $\mathbf{R}_{ss} + \mathbf{R}_{ww}$ and the fact that $\mathbf{h}$ is just $\mathbf{a}$ when flipped upside down. Written out, the set of linear equations becomes

$$\begin{bmatrix} r_{xx}[0] & r_{xx}[1] & \cdots & r_{xx}[n] \\ r_{xx}[1] & r_{xx}[0] & \cdots & r_{xx}[n-1] \\ \vdots & \vdots & \ddots & \vdots \\ r_{xx}[n] & r_{xx}[n-1] & \cdots & r_{xx}[0] \end{bmatrix} \begin{bmatrix} h^{(n)}[0] \\ h^{(n)}[1] \\ \vdots \\ h^{(n)}[n] \end{bmatrix} = \begin{bmatrix} r_{ss}[0] \\ r_{ss}[1] \\ \vdots \\ r_{ss}[n] \end{bmatrix} \tag{12.59}$$

where $r_{xx}[k] = r_{ss}[k] + r_{ww}[k]$. These are the *Wiener-Hopf filtering equations.* It appears that they must be solved for each value of n. However, a computationally efficient algorithm for doing so is the Levinson recursion, which solves the equations recursively to avoid resolving them for each value of n [Marple 1987]. For large enough

n it can be shown that the filter becomes time invariant so that only a single solution is necessary. In this case an analytical solution may be possible. The Wiener-Hopf filtering equations can be written as

$$\sum_{k=0}^{n} h^{(n)}[k] r_{xx}[l-k] = r_{ss}[l] \qquad l = 0, 1, \ldots, n$$

where we have used the property $r_{xx}[-k] = r_{xx}[k]$. As $n \to \infty$, we have upon replacing the time varying impulse response $h^{(n)}[k]$ by its time invariant version $h[k]$

$$\sum_{k=0}^{\infty} h[k] r_{xx}[l-k] = r_{ss}[l] \qquad l = 0, 1, \ldots \; . \tag{12.60}$$

The same set of equations result if we attempt to estimate $s[n]$ based on the present and infinite past or based on $x[m]$ for $m \leq n$. This is termed the *infinite Wiener filter*. To see why this is so we let

$$\hat{s}[n] = \sum_{k=0}^{\infty} h[k] x[n-k]$$

and use the orthogonality principle (12.13). Then, we have

$$s[n] - \hat{s}[n] \perp \ldots, x[n-1], x[n]$$

or by using the definition of orthogonality

$$E\left[(s[n] - \hat{s}[n]) x[n-l]\right] = 0 \qquad l = 0, 1, \ldots \; .$$

Hence,

$$E\left(\sum_{k=0}^{\infty} h[k] x[n-k] x[n-l]\right) = E(s[n] x[n-l])$$

and therefore, the equations to be solved for the infinite Wiener filter impulse response are

$$\sum_{k=0}^{\infty} h[k] r_{xx}[l-k] = r_{ss}[l] \qquad l = 0, 1, \ldots \; .$$

A little thought will convince the reader that the solutions must be identical since the problem of estimating $s[n]$ based on $x[m]$ for $0 \leq m \leq n$ as $n \to \infty$ is really just that of using the present and infinite past to estimate the current sample. The time invariance of the filter makes the solution independent of which sample is to be estimated or independent of n. The solution of (12.60) utilizes spectral factorization and is explored in Problem 12.16. At first glance it might appear that (12.60) could be solved by using Fourier transform techniques since the left-hand side of the equation is a convolution of two sequences. This is not the case, however, since the equations hold only for $l \geq 0$. If, indeed, they were valid for $l < 0$ as well, then the Fourier transform approach would be viable. Such a set of equations arises in the smoothing problem in which $s[n]$ is

to be estimated based on $\{\ldots, x[-1], x[0], x[1], \ldots\}$ or $x[k]$ for *all* k. In this case the smoothing estimator takes the form

$$\hat{s}[n] = \sum_{k=-\infty}^{\infty} a_k x[k]$$

and by letting $h[k] = a_{n-k}$ we have the convolution sum

$$\hat{s}[n] = \sum_{k=-\infty}^{\infty} h[k]x[n-k]$$

where $h[k]$ is the impulse response of an infinite two-sided time invariant filter. The Wiener-Hopf equations become

$$\sum_{k=-\infty}^{\infty} h[k]r_{xx}[l-k] = r_{ss}[k] \qquad -\infty < l < \infty \tag{12.61}$$

(see Problem 12.17). The difference from the filtering case is that now the equations must be satisfied for all l, and there is no constraint that $h[k]$ must be causal. Hence, we can use Fourier transform techniques to solve for the impulse response since (12.61) becomes

$$h[n] \star r_{xx}[n] = r_{ss}[n] \tag{12.62}$$

where $\star$ denotes convolution. Letting $H(f)$ be the frequency response of the infinite Wiener smoother, we have upon taking the Fourier transform of (12.62):

$$\begin{aligned} H(f) &= \frac{P_{ss}(f)}{P_{xx}(f)} \\ &= \frac{P_{ss}(f)}{P_{ss}(f) + P_{ww}(f)}. \end{aligned}$$

The frequency response is a real even function of frequency, so that the impulse response must also be real and even. This means that the filter is not causal. Of course, this is to be expected since we sought to estimate $s[n]$ using *future* as well as present and past data. Before leaving this topic it is worthwhile to point out that the Wiener smoother emphasizes portions of the frequency spectrum of the data where the SNR is high and attenuates those where it is low. This is evident if we define the "local SNR" as the SNR in a narrow band of frequencies centered about f as

$$\eta(f) = \frac{P_{ss}(f)}{P_{ww}(f)}.$$

Then, the optimal filter frequency response becomes

$$H(f) = \frac{\eta(f)}{\eta(f)+1}.$$

Clearly, the filter response satisfies $0 < H(f) < 1$, and thus the Wiener smoother response is $H(f) \approx 0$ when $\eta(f) \approx 0$ (low local SNR) and $H(f) \approx 1$ when $\eta(f) \to \infty$ (high local SNR). The reader may wish to compare these results to those of the Wiener filter given by (12.56). See also Problem 12.18 for further results on the infinite Wiener smoother.

Finally, we examine the prediction problem in which we wish to estimate $\theta = x[N-1+l]$ for $l \geq 1$ based on $\mathbf{x} = [x[0]\, x[1] \ldots x[N-1]]^T$. The resulting estimator is termed the *l-step linear predictor*. We use (12.51) for which

$$\mathbf{C}_{xx} = \mathbf{R}_{xx}$$

where $\mathbf{R}_{xx}$ is of dimension $N \times N$ and

$$\begin{aligned} \mathbf{C}_{\theta x} &= E\left[x[N-1+l]\left[\begin{array}{cccc} x[0] & x[1] & \ldots & x[N-1] \end{array}\right]\right] \\ &= \left[\begin{array}{cccc} r_{xx}[N-1+l] & r_{xx}[N-2+l] & \ldots & r_{xx}[l] \end{array}\right]. \end{aligned}$$

Let the latter vector be denoted by $\mathbf{r}'^T_{xx}$. Then,

$$\hat{x}[N-1+l] = \mathbf{r}'^T_{xx}\mathbf{R}^{-1}_{xx}\mathbf{x}.$$

Recalling that

$$\mathbf{a} = \mathbf{R}^{-1}_{xx}\mathbf{r}'_{xx} \tag{12.63}$$

we have

$$\hat{x}[N-1+l] = \sum_{k=0}^{N-1} a_k x[k].$$

If we let $h[N-k] = a_k$ to allow a "filtering" interpretation, then

$$\begin{aligned} \hat{x}[N-1+l] &= \sum_{k=0}^{N-1} h[N-k]x[k] \\ &= \sum_{k=1}^{N} h[k]x[N-k] \end{aligned} \tag{12.64}$$

and it is observed that the predicted sample is the output of a filter with impulse response $h[n]$. The equations to be solved are from (12.63) (noting once again that $\mathbf{h}$ is just $\mathbf{a}$ when flipped upside down)

$$\mathbf{R}_{xx}\mathbf{h} = \mathbf{r}_{xx}$$

where $\mathbf{r}_{xx} = [r_{xx}[l]\, r_{xx}[l+1] \ldots r_{xx}[N-1+l]]^T$. In explicit form they become

$$\left[\begin{array}{cccc} r_{xx}[0] & r_{xx}[1] & \ldots & r_{xx}[N-1] \\ r_{xx}[1] & r_{xx}[0] & \ldots & r_{xx}[N-2] \\ \vdots & \vdots & \ddots & \vdots \\ r_{xx}[N-1] & r_{xx}[N-2] & \ldots & r_{xx}[0] \end{array}\right] \left[\begin{array}{c} h[1] \\ h[2] \\ \vdots \\ h[N] \end{array}\right]$$

$$= \begin{bmatrix} r_{xx}[l] \\ r_{xx}[l+1] \\ \vdots \\ r_{xx}[N-1+l] \end{bmatrix}. \tag{12.65}$$

These are the Wiener-Hopf prediction equations for the l-step linear predictor based on N past samples. A computationally efficient method for solving these equations is the Levinson recursion [Marple 1987]. For the specific case where $l = 1$, the one-step linear predictor, the values of $-h[n]$ are termed the *linear prediction coefficients* which are used extensively in speech modeling [Makhoul 1975]. Also, for $l = 1$ the resulting equations are identical to the Yule-Walker equations used to solve for the AR filter parameters of an AR(N) process (see Example 7.18 and Appendix 1).

The minimum MSE for the l-step linear predictor is, from (12.52),

$$M_{\hat{x}} = r_{xx}[0] - \mathbf{r}_{xx}'^T \mathbf{R}_{xx}^{-1} \mathbf{r}_{xx}'$$

or, equivalently,

$$\begin{aligned} M_{\hat{x}} &= r_{xx}[0] - \mathbf{r}_{xx}'^T \mathbf{a} \\ &= r_{xx}[0] - \sum_{k=0}^{N-1} a_k r_{xx}[N-1+l-k] \\ &= r_{xx}[0] - \sum_{k=0}^{N-1} h[N-k] r_{xx}[N-1+l-k] \\ &= r_{xx}[0] - \sum_{k=1}^{N} h[k] r_{xx}[k+(l-1)]. \end{aligned} \tag{12.66}$$

As an example, assume that $x[n]$ is an AR(1) process with ACF (see Appendix 1)

$$r_{xx}[k] = \frac{\sigma_u^2}{1-a^2[1]} \left(-a[1]\right)^{|k|}$$

and we wish to find the one-step predictor $\hat{x}[N]$ as (see (12.64))

$$\hat{x}[N] = \sum_{k=1}^{N} h[k] x[N-k].$$

To solve for the $h[k]$'s we use (12.65) for $l = 1$, which is

$$\sum_{k=1}^{N} h[k] r_{xx}[m-k] = r_{xx}[m] \qquad m = 1, 2, \ldots, N.$$

Substituting the ACF, we have

$$\sum_{k=1}^{N} h[k] \left(-a[1]\right)^{|m-k|} = \left(-a[1]\right)^{|m|}.$$

It is easily verified that these equations are solved for

$$h[k] = \begin{cases} -a[1] & k = 1 \\ 0 & k = 2, 3, \ldots, N. \end{cases}$$

Hence, the one-step linear predictor is

$$\hat{x}[N] = -a[1]x[N-1]$$

and depends only on the previous sample. This is not surprising in light of the fact that the AR(1) process satisfies

$$x[n] = -a[1]x[n-1] + u[n]$$

where $u[n]$ is white noise, so that the sample to be predicted satisfies

$$x[N] = -a[1]x[N-1] + u[N].$$

The predictor cannot predict $u[N]$ since it is uncorrelated with the past data samples (recall that since the AR(1) filter is causal, $x[n]$ is a linear combination of $\{u[n], u[n-1], \ldots\}$ and thus $x[n]$ is uncorrelated with all future samples of $u[n]$). The prediction error is $x[N] - \hat{x}[N] = u[N]$, and the minimum MSE is just the driving noise power σ_u^2. To verify this we have from (12.66)

$$\begin{aligned} M_{\hat{x}} &= r_{xx}[0] - \sum_{k=1}^{N} h[k] r_{xx}[k] \\ &= r_{xx}[0] + a[1] r_{xx}[1] \\ &= \frac{\sigma_u^2}{1 - a^2[1]} + a[1] \frac{\sigma_u^2}{1 - a^2[1]} (-a[1]) \\ &= \sigma_u^2. \end{aligned}$$

We can extend these results to the l-step predictor by solving

$$\sum_{k=1}^{N} h[k] r_{xx}[m-k] = r_{xx}[m+l-1] \qquad m = 1, 2, \ldots, N.$$

Substituting the ACF for an AR(1) process this becomes

$$\sum_{k=1}^{N} h[k] \left(-a[1]\right)^{|m-k|} = \left(-a[1]\right)^{|m+l-1|}.$$

The solution, which can be easily verified, is

$$h[k] = \begin{cases} \left(-a[1]\right)^{l} & k = 1 \\ 0 & k = 2, 3, \ldots, N \end{cases}$$

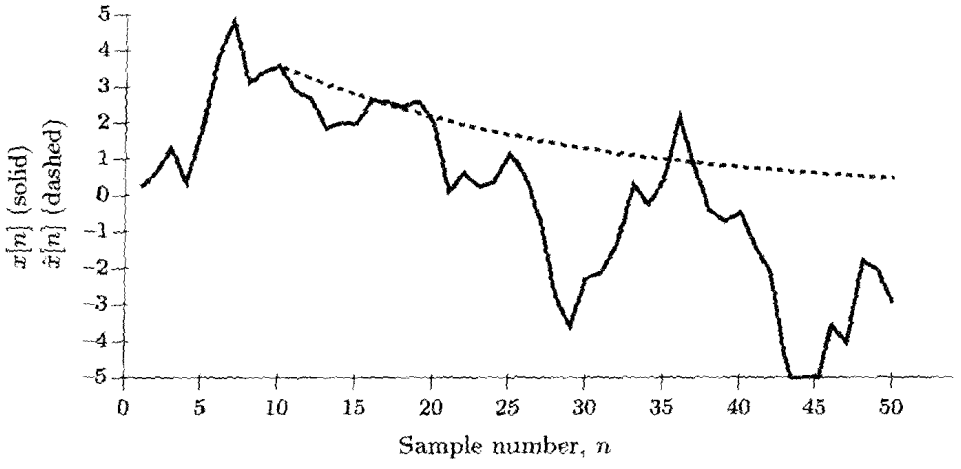


Figure 12.8 Linear prediction for realization of AR(1) process

and therefore the l-step predictor is

$$\hat{x}[(N-1)+l] = (-a[1])^l x[N-1]. \tag{12.67}$$

The minimum MSE is, from (12.66),

$$\begin{aligned} M_{\hat{x}} &= r_{xx}[0] - h[1] r_{xx}[l] \\ &= \frac{\sigma_u^2}{1-a^2[1]} - (-a[1])^l \frac{\sigma_u^2}{1-a^2[1]} (-a[1])^l \\ &= \frac{\sigma_u^2}{1-a^2[1]} \left(1 - a^{2l}[1]\right). \end{aligned}$$

It is interesting to note that the predictor decays to zero with l (since $|a[1]| < 1$). This is reasonable since the correlation between $x[(N-1)+l]$, the sample to be predicted, and $x[N-1]$, the data sample on which the prediction is based, is $r_{xx}[l]$. As l increases, $r_{xx}[l]$ decays to zero and thus so does $\hat{x}[(N-1)+l]$. This is also reflected in the minimum MSE, which is smallest for $l = 1$ and increases for larger l.

A numerical example illustrates the behavior of the l-step predictor. If $a[1] = -0.95$ and $\sigma_u^2 = 0.1$, so that the process has a low-pass PSD, then for a given realization of $x[n]$ we obtain the results shown in Figure 12.8. The true data are displayed as a solid line, while the predicted data for $n \geq 11$ is shown as a dashed line. The predictions are given by (12.67), where $N = 11$ and $l = 1, 2, \ldots, 40$, and thus decay to zero with increasing l. As can be observed, the predictions are generally poor except for small l. See also Problems 12.19 and 12.20.

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Problems

12.1 Consider the quadratic estimator

$$\hat{\theta} = ax^2[0] + bx[0] + c$$

of a scalar parameter θ based on the single data sample $x[0]$. Find the coefficients a, b, c that minimize the Bayesian MSE. If $x[0] \sim \mathcal{U}[-\frac{1}{2}, \frac{1}{2}]$, find the LMMSE estimator and the quadratic MMSE estimator if $\theta = \cos 2\pi x[0]$. Also, compare the minimum MSEs.

12.2 Consider the data

$$x[n] = Ar^n + w[n] \quad n = 0, 1, \ldots, N-1$$

where A is a parameter to be estimated, r is a known constant, and $w[n]$ is zero mean white noise with variance σ^2. The parameter A is modeled as a random variable with mean μ_A and variance σ_A^2 and is independent of $w[n]$. Find the LMMSE estimator of A and the minimum Bayesian MSE.

12.3 A Gaussian random vector $\mathbf{x} = [x_1 \; x_2]^T$ has zero mean and covariance matrix $\mathbf{C}_{xx}$. If x_2 is to be linearly estimated based on x_1, find the estimator that minimizes the Bayesian MSE. Also, find the minimum MSE and prove that it is zero if and only if $\mathbf{C}_{xx}$ is singular. Extend your results to show that if the covariance matrix of an $N \times 1$ zero mean Gaussian random vector is not positive definite, then any random variable may be perfectly estimated by a linear combination of the others. Hint: Note that

$$E\left[\left(\sum_{i=1}^{N} a_i x_i\right)^2\right] = \mathbf{a}^T \mathbf{C}_{xx} \mathbf{a}.$$

12.4 An inner product (x, y) between two vectors x and y of a vector space must satisfy the following properties

a. $(x, x) \geq 0$ and $(x, x) = 0$ if and only if $x = 0$.

b. $(x, y) = (y, x)$.

c. $(c_1 x_1 + c_2 x_2, y) = c_1(x_1, y) + c_2(x_2, y)$.

Prove that the definition $(x, y) = E(xy)$ for x and y zero mean random variables satisfies these properties.

12.5 If we assume nonzero mean random variables, then a reasonable approach is to define the inner product between x and y as $(x, y) = \text{cov}(x, y)$. With this definition x and y are orthogonal if and only if they are uncorrelated. For this "inner product" which of the properties given in Problem 12.4 is violated?

12.6 We observe the data $x[n] = s[n] + w[n]$ for $n = 0, 1, \ldots, N-1$, where $s[n]$ and $w[n]$ are zero mean WSS random processes which are uncorrelated with each other. The ACFs are

$$\begin{aligned} r_{ss}[k] &= \sigma_s^2 \delta[k] \\ r_{ww}[k] &= \sigma^2 \delta[k]. \end{aligned}$$

Determine the LMMSE estimator of $\mathbf{s} = [s[0]\, s[1] \ldots s[N-1]]^T$ based on $\mathbf{x} = [x[0]\, x[1] \ldots x[N-1]]^T$ and the corresponding minimum MSE matrix.

12.7 Derive the Bayesian MSE matrix for the vector LMMSE estimator as given by (12.21), as well as the minimum Bayesian MSE of (12.22). Keep in mind the averaging PDFs implied by the expectation operator.

12.8 Prove the commutative and additive properties of the LMMSE estimator as given by (12.23) and (12.24).

12.9 Derive a sequential LMMSE estimator analogous to (12.34)–(12.36) but for the case where $\mu_A \neq 0$. Hint: Use an algebraic approach based on the results in Example 10.1.

12.10 Given a set of vectors $\{x_1, x_2, \ldots, x_n\}$, the Gram-Schmidt orthogonalization procedure finds a new set of vectors $\{e_1, e_2, \ldots, e_n\}$ which are orthonormal (orthogonal and having unit length) or $(e_i, e_j) = \delta_{ij}$. The procedure is

a. $e_1 = \dfrac{x_1}{||x_1||}$

b. $z_2 = x_2 - (x_2, e_1)e_1$

$e_2 = \dfrac{z_2}{||z_2||}$

c. and so on,

or in general for $n \geq 2$

$$\begin{aligned} z_n &= x_n - \sum_{i=1}^{n-1} (x_n, e_i) e_i \\ e_n &= \frac{z_n}{||z_n||}. \end{aligned}$$

Give a geometrical interpretation of the procedure. For the Euclidean vectors

$$\begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}, \quad \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \quad \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

find three orthonormal vectors using the Gram-Schmidt procedure. The inner product is defined as $(\mathbf{x}, \mathbf{y}) = \mathbf{x}^T\mathbf{y}$ for $\mathbf{x}$ and $\mathbf{y}$ Euclidean vectors.

12.11 Let $\mathbf{x}$ denote the vector composed of three zero mean random variables with a covariance matrix

$$\mathbf{C}_{xx} = \begin{bmatrix} 1 & \rho & \rho^2 \\ \rho & 1 & \rho \\ \rho^2 & \rho & 1 \end{bmatrix}.$$

If $\mathbf{y} = \mathbf{A}\mathbf{x}$, determine the 3×3 matrix $\mathbf{A}$ so that the covariance matrix of $\mathbf{y}$ is $\mathbf{I}$ or, equivalently, so that the random variables $\{y_1, y_2, y_3\}$ are uncorrelated and have unit variance. Use the Gram-Schmidt orthogonalization procedure in Problem 12.10 to do so. What is interesting about $\mathbf{A}$? Finally, relate $\mathbf{C}_{xx}^{-1}$ to $\mathbf{A}$. Note that $\mathbf{A}$ may be viewed as a *whitening* transformation (see Section 4.5).

12.12 For the sequential LMMSE estimator of a scalar parameter explain what would happen if $\sigma_n^2 \to 0$ for some n. Do you obtain the same results for the vector case?

12.13 Find the sequential LMMSE estimator for the problem described in Example 12.1. Verify that the sequential estimator is identical to the batch estimator (12.9). Hint: Solve for the $1/K[n]$ sequence.

12.14 In this problem we examine the interpolation of a data sample. We assume that the data set $\{x[n-M], \ldots, x[n-1], x[n+1], \ldots, x[n+M]\}$ is available and that we wish to estimate or interpolate $x[n]$. The data and $x[n]$ are assumed to be a realization of a zero mean WSS random process. Let the LMMSE estimator of $x[n]$ be

$$\hat{x}[n] = \sum_{\substack{k=-M \\ k \neq 0}}^{M} a_k x[n-k].$$

Find the set of linear equations to be solved for the weighting coefficients by using the orthogonality principle. Next, prove that $a_{-k} = a_k$ and explain why this must be true. See also [Kay 1983] for a further discussion of interpolation.

12.15 Consider the Wiener smoother for a single data sample as given by (12.56). Rewrite W and $M_{\hat{s}}$ as a function of the correlation coefficient

$$\rho = \frac{\text{cov}(s[0], x[0])}{\sqrt{\text{var}(s[0])\text{var}(x[0])}}$$

between $s[0]$ and $x[0]$. Explain your results.

12.16 In this problem we explore the solution of the Wiener-Hopf filter equations based on the present and infinite past or the solution of

$$\sum_{k=0}^{\infty} h[k] r_{xx}[l-k] = r_{ss}[l] \qquad l \geq 0$$

by the use of z transforms. First we define the one-sided z transform as

$$\begin{aligned} [\mathcal{X}(z)]_+ &= \left[\sum_{n=-\infty}^{\infty} x[n] z^{-n}\right]_+ \\ &= \sum_{n=0}^{\infty} x[n] z^{-n}. \end{aligned}$$

This is seen to be the usual z transform but with the positive powers of z omitted. Next, we write the Wiener-Hopf equation as

$$h[n] \star r_{xx}[n] - r_{ss}[n] = 0 \qquad n \geq 0 \tag{12.68}$$

where $h[n]$ is constrained to be causal. The z transform (two-sided) of the left-hand side is

$$\mathcal{H}(z)\mathcal{P}_{xx}(z) - \mathcal{P}_{ss}(z)$$

so that to satisfy (12.68) we must have

$$[\mathcal{H}(z)\mathcal{P}_{xx}(z) - \mathcal{P}_{ss}(z)]_+ = 0.$$

By the spectral factorization theorem [Orfanidis 1985], if $\mathcal{P}_{xx}(z)$ has no zeros on the unit circle, then it may be factored as

$$\mathcal{P}_{xx}(z) = \mathcal{B}(z)\mathcal{B}(z^{-1})$$

where $\mathcal{B}(z)$ is the z transform of a causal sequence and $\mathcal{B}(z^{-1})$ is the z transform of an anticausal sequence. Hence,

$$[\mathcal{H}(z)\mathcal{B}(z)\mathcal{B}(z^{-1}) - \mathcal{P}_{ss}(z)]_+ = 0.$$

Let $\mathcal{G}(z) = \mathcal{H}(z)\mathcal{B}(z)$, so that

$$\left[\mathcal{B}(z^{-1})\left(\mathcal{G}(z) - \frac{\mathcal{P}_{ss}(z)}{\mathcal{B}(z^{-1})}\right)\right]_+ = 0.$$

Noting that $\mathcal{G}(z)$ is the z transform of a causal sequence, show that

$$\mathcal{H}(z) = \frac{1}{\mathcal{B}(z)}\left[\frac{\mathcal{P}_{ss}(z)}{\mathcal{B}(z^{-1})}\right]_+.$$

For an example of the computations involved in determining the Wiener filter, see [Orfanidis 1985].

12.17 Rederive the infinite Wiener smoother (12.61) by first assuming that

$$\hat{s}[n] = \sum_{k=-\infty}^{\infty} h[k]x[n-k]$$

and then using the orthogonality principle.

12.18 For the infinite Wiener smoother show that the minimum Bayesian MSE is

$$M_{\hat{s}} = r_{ss}[0] - \sum_{k=-\infty}^{\infty} h[k]r_{ss}[k].$$

Then, using Fourier transform techniques, show that this can be rewritten as

$$M_{\hat{s}} = \int_{-\frac{1}{2}}^{\frac{1}{2}} P_{ss}(f)\,(1 - H(f))\,df$$

where

$$H(f) = \frac{P_{ss}(f)}{P_{ss}(f) + P_{ww}(f)}.$$

Evaluate the Wiener smoother and minimum MSE if

$$\begin{aligned} P_{ss}(f) &= \begin{cases} P_0 & |f| \leq \frac{1}{4} \\ 0 & \frac{1}{4} < |f| \leq \frac{1}{2} \end{cases} \\ P_{ww}(f) &= \sigma^2 \end{aligned}$$

and explain your results.

12.19 Assume that $x[n]$ is a zero mean WSS random process. We wish to predict $x[n]$ based on $\{x[n-1], x[n-2] \ldots, x[n-N]\}$. Use the orthogonality principle to derive the LMMSE estimator or predictor. Explain why the equations to be solved are the same as (12.65) for $l = 1$, i.e., they are independent of n. Also, rederive the minimum MSE (12.66) by again invoking the orthogonality principle.

12.20 Consider an AR(N) process

$$x[n] = -\sum_{k=1}^{N} a[k]x[n-k] + u[n]$$

where $u[n]$ is white noise with variance σ_u^2. Prove that the optimal one-step linear predictor of $x[n]$ is

$$\hat{x}[n] = -\sum_{k=1}^{N} a[k]x[n-k].$$

Also, find the minimum MSE. Hint: Compare the equations to be solved to the Yule-Walker equations (see Appendix 1).

Appendix 12A

Derivation of Sequential LMMSE Estimator for the Bayesian Linear Model

We will use a vector space approach to update the ith component of $\hat{\boldsymbol{\theta}}[n-1]$ as follows.

$$\hat{\theta}_i[n] = \hat{\theta}_i[n-1] + K_i[n]\left(x[n] - \hat{x}[n|n-1]\right) \tag{12A.1}$$

where $\hat{x}[n|n-1]$ is the LMMSE estimator of $x[n]$ based on $\{x[0], x[1], \ldots, x[n-1]\}$. We now shorten the notation from that used in Section 12.6 where we denoted the same LMMSE estimator as $\hat{x}[n|0, 1, \ldots, n-1]$. The motivation for (12A.1) follows by the same rationale as depicted in Figure 12.5 (see also (12.41)). Also, we let $\tilde{x}[n] = x[n] - \hat{x}[n|n-1]$, which is the innovation sequence. We can then form the update for the entire parameter as

$$\hat{\boldsymbol{\theta}}[n] = \hat{\boldsymbol{\theta}}[n-1] + \mathbf{K}[n](x[n] - \hat{x}[n|n-1]).$$

Before beginning the derivation we state some properties which are needed. The reader should recall that all random variables are assumed to be zero mean.

1. The LMMSE estimator of $\mathbf{A}\boldsymbol{\theta}$ is $\mathbf{A}\hat{\boldsymbol{\theta}}$, the commutative property of (12.23).

2. The LMMSE estimator of $\boldsymbol{\theta}_1 + \boldsymbol{\theta}_2$ is $\hat{\boldsymbol{\theta}}_1 + \hat{\boldsymbol{\theta}}_2$, the additive property of (12.24).

3. Since $\hat{\theta}_i[n-1]$ is a linear combination of the samples $\{x[0], x[1], \ldots, x[n-1]\}$, and the innovation $x[n] - \hat{x}[n|n-1]$ is uncorrelated with the past data samples, it follows that

$$E\left[\hat{\theta}_i[n-1]\left(x[n] - \hat{x}[n|n-1]\right)\right] = 0.$$

4. Since $\boldsymbol{\theta}$ and $w[n]$ are uncorrelated as assumed in the Bayesian linear model and $\hat{\boldsymbol{\theta}}[n-1]$ and $w[n]$ are also uncorrelated, it follows that

$$E\left[\left(\boldsymbol{\theta} - \hat{\boldsymbol{\theta}}[n-1]\right)w[n]\right] = \mathbf{0}.$$

To see why $\hat{\boldsymbol{\theta}}[n-1]$ and $w[n]$ are uncorrelated note first that $\hat{\boldsymbol{\theta}}[n-1]$ depends linearly on the past data samples or on $\boldsymbol{\theta}$ and $\{w[0], w[1], \ldots, w[n-1]\}$. But $w[n]$ is uncorrelated with $\boldsymbol{\theta}$ and by the assumption that $\mathbf{C}_w$ is *diagonal* or that $w[n]$ is a sequence of uncorrelated random variables, $w[n]$ is uncorrelated with the past noise samples.

With these results the derivation simplifies. To begin we note that

$$x[n] = \mathbf{h}^T[n]\boldsymbol{\theta} + w[n].$$

Using properties 1 and 2 we have that

$$\hat{x}[n|n-1] = \mathbf{h}^T[n]\hat{\boldsymbol{\theta}}[n-1] + \hat{w}[n|n-1].$$

But $\hat{w}[n|n-1] = 0$ since $w[n]$ is uncorrelated with the past data samples, as explained in property 4. Thus,

$$\hat{x}[n|n-1] = \mathbf{h}^T[n]\hat{\boldsymbol{\theta}}[n-1].$$

We next find the gain factor $K_i[n]$. Using (12.37), we have

$$K_i[n] = \frac{E\left[\theta_i(x[n] - \hat{x}[n|n-1])\right]}{E\left[\left(x[n] - \hat{x}[n|n-1]\right)^2\right]}. \tag{12A.2}$$

Evaluating the denominator

$$\begin{aligned} E\left[\left(x[n] - \hat{x}[n|n-1]\right)^2\right] &= E\left[\left(\mathbf{h}^T[n]\boldsymbol{\theta} + w[n] - \mathbf{h}^T[n]\hat{\boldsymbol{\theta}}[n-1]\right)^2\right] \\ &= E\left[\left(\mathbf{h}^T[n](\boldsymbol{\theta} - \hat{\boldsymbol{\theta}}[n-1]) + w[n]\right)^2\right] \\ &= \mathbf{h}^T[n]E\left[(\boldsymbol{\theta} - \hat{\boldsymbol{\theta}}[n-1])(\boldsymbol{\theta} - \hat{\boldsymbol{\theta}}[n-1])^T\right]\mathbf{h}[n] \\ &\quad + E(w^2[n]) \\ &= \mathbf{h}^T[n]\mathbf{M}[n-1]\mathbf{h}[n] + \sigma_n^2 \end{aligned}$$

where we have used property 4. Now, using properties 3 and 4 to evaluate the numerator

$$\begin{aligned} E\left[\theta_i(x[n] - \hat{x}[n|n-1])\right] &= E\left[(\theta_i - \hat{\theta}_i[n-1])(x[n] - \hat{x}[n|n-1])\right] \\ &= E\left[(\theta_i - \hat{\theta}_i[n-1])(\mathbf{h}^T[n]\boldsymbol{\theta} + w[n] - \mathbf{h}^T[n]\hat{\boldsymbol{\theta}}[n-1])\right] \\ &= E\left[(\theta_i - \hat{\theta}_i[n-1])\left(\mathbf{h}^T[n](\boldsymbol{\theta} - \hat{\boldsymbol{\theta}}[n-1])\right)\right] \\ &= E\left[(\theta_i - \hat{\theta}_i[n-1])(\boldsymbol{\theta} - \hat{\boldsymbol{\theta}}[n-1])^T\right]\mathbf{h}[n] \end{aligned} \tag{12A.3}$$

so that

$$K_i[n] = \frac{E\left[(\theta_i - \hat{\theta}_i[n-1])(\boldsymbol{\theta} - \hat{\boldsymbol{\theta}}[n-1])^T\right]\mathbf{h}[n]}{\mathbf{h}^T[n]\mathbf{M}[n-1]\mathbf{h}[n] + \sigma_n^2}$$

and therefore

$$\mathbf{K}[n] = \frac{\mathbf{M}[n-1]\mathbf{h}[n]}{\mathbf{h}^T[n]\mathbf{M}[n-1]\mathbf{h}[n] + \sigma_n^2}. \tag{12A.4}$$

To determine the update for the MSE matrix

$$\begin{aligned}
\mathbf{M}[n] &= E\left[(\boldsymbol{\theta} - \hat{\boldsymbol{\theta}}[n])(\boldsymbol{\theta} - \hat{\boldsymbol{\theta}}[n])^T\right] \\
&= E\left[\left(\boldsymbol{\theta} - \hat{\boldsymbol{\theta}}[n-1] - \mathbf{K}[n](x[n] - \hat{x}[n|n-1])\right)\right. \\
&\quad \left.\cdot \left(\boldsymbol{\theta} - \hat{\boldsymbol{\theta}}[n-1] - \mathbf{K}[n](x[n] - \hat{x}[n|n-1])\right)^T\right] \\
&= \mathbf{M}[n-1] - E\left[(\boldsymbol{\theta} - \hat{\boldsymbol{\theta}}[n-1])(x[n] - \hat{x}[n|n-1])\mathbf{K}^T[n]\right] \\
&\quad - \mathbf{K}[n]E\left[(x[n] - \hat{x}[n|n-1])(\boldsymbol{\theta} - \hat{\boldsymbol{\theta}}[n-1])^T\right] \\
&\quad + \mathbf{K}[n]E\left[(x[n] - \hat{x}[n|n-1])^2\right]\mathbf{K}^T[n].
\end{aligned}$$

But from (12A.2) and (12A.4)

$$\mathbf{K}[n]E\left[(x[n] - \hat{x}[n|n-1])^2\right] = \mathbf{M}[n-1]\mathbf{h}[n]$$

and from property 3 and (12A.3)

$$\begin{aligned}
E\left[(\boldsymbol{\theta} - \hat{\boldsymbol{\theta}}[n-1])(x[n] - \hat{x}[n|n-1])\right] &= E\left[\boldsymbol{\theta}(x[n] - \hat{x}[n|n-1])\right] \\
&= \mathbf{M}[n-1]\mathbf{h}[n]
\end{aligned}$$

so that

$$\begin{aligned}
\mathbf{M}[n] &= \mathbf{M}[n-1] - \mathbf{M}[n-1]\mathbf{h}[n]\mathbf{K}^T[n] - \mathbf{K}[n]\mathbf{h}^T[n]\mathbf{M}[n-1] \\
&\quad + \mathbf{M}[n-1]\mathbf{h}[n]\mathbf{K}^T[n] \\
&= \left(\mathbf{I} - \mathbf{K}[n]\mathbf{h}^T[n]\right)\mathbf{M}[n-1].
\end{aligned}$$

We next show that the same equations result if $\boldsymbol{\theta}$ and $x[n]$ are not zero mean. Since the sequential implementation of the minimum MSE matrix (12.49) must be identical to (12.28), which does not depend on the means, $\mathbf{M}[n]$ likewise must be independent of the means. Also, since $\mathbf{M}[n]$ depends on $\mathbf{K}[n]$, the gain vector must also be independent of the means. Finally, the estimator update equation (12.47) is valid for zero means, as we have already shown in this appendix. If the means are not zero, we may still apply (12.47) to the data set $x[n] - E(x[n])$ and view the estimate as that of $\boldsymbol{\theta} - E(\boldsymbol{\theta})$. By the commutative property, however, the LMMSE estimator of $\boldsymbol{\theta} + \mathbf{b}$ for $\mathbf{b}$ a constant vector is $\hat{\boldsymbol{\theta}} + \mathbf{b}$ (see (12.23)). Thus, (12.47) becomes

$$\hat{\boldsymbol{\theta}}[n] - E(\boldsymbol{\theta}) = \hat{\boldsymbol{\theta}}[n-1] - E(\boldsymbol{\theta}) + \mathbf{K}[n]\left[x[n] - E(x[n]) - \mathbf{h}^T[n](\hat{\boldsymbol{\theta}}[n-1] - E(\boldsymbol{\theta}))\right]$$

where $\hat{\boldsymbol{\theta}}[n]$ is the LMMSE estimator for *nonzero means*. Rearranging and canceling terms we have

$$\hat{\boldsymbol{\theta}}[n] = \hat{\boldsymbol{\theta}}[n-1] + \mathbf{K}[n]\left[x[n] - \mathbf{h}^T[n]\hat{\boldsymbol{\theta}}[n-1] - (E(x[n]) - \mathbf{h}^T[n]E(\boldsymbol{\theta}))\right]$$

and since

$$\begin{aligned} E(x[n]) &= E\left(\mathbf{h}^T[n]\boldsymbol{\theta} + w[n]\right) \\ &= \mathbf{h}^T[n]E(\boldsymbol{\theta}) \end{aligned}$$

we arrive at the identical equation for the estimator update.